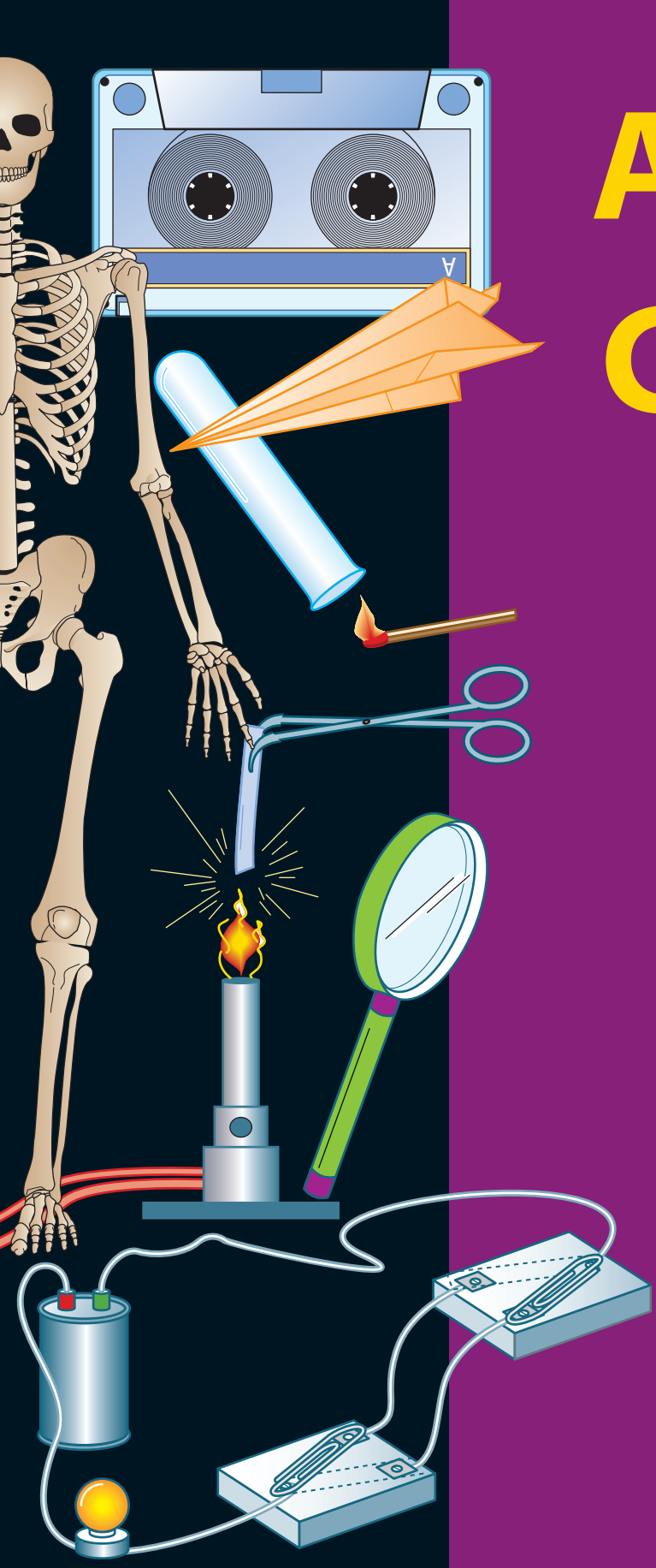

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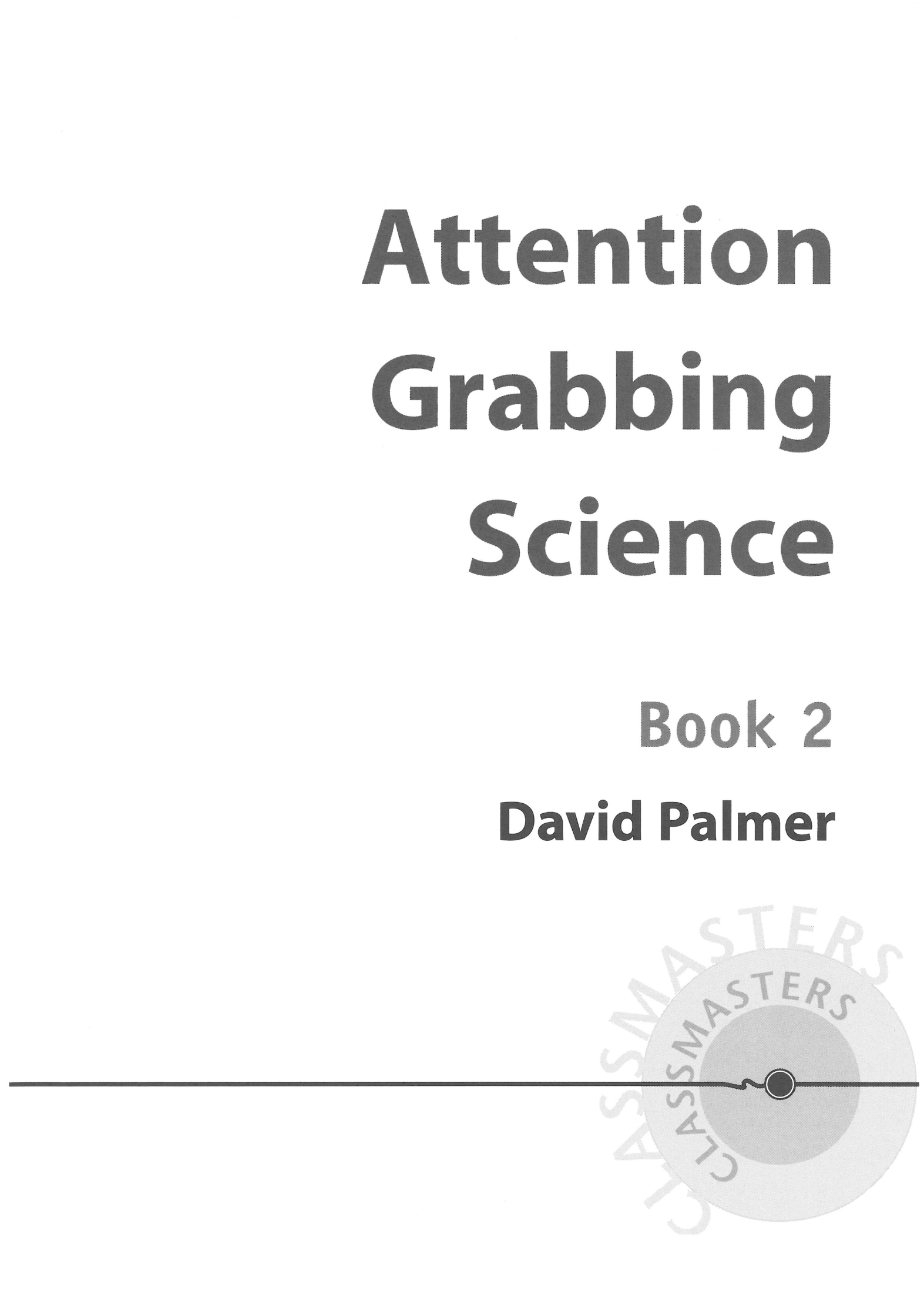

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This book is dedicated to the late Terry Lear.

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PREFACE

Professor Julius Sumner Miller showed that science activities that have the element of surprise (that is, an unexpected result) are interesting to people of all ages. A surprising result can motivate students to examine their own ideas and to want to know 'Why is it so?'. With similar thinking, I decided to base this book on a collection of surprising science activities.

However, learning science is not just about whiz-bang activities, so this book also contains lots of suggestions for other types of motivational science experiences. These have been grouped together into themes so that students can experience a variety of activities in relation to each concept. The 'To Investigate' sections have ideas for student-designed investigations. The shaded boxes contain fascinating facts chosen for their surprise value, as well as interesting historical anecdotes. The 'Linked Activities' sections contain further surprising activities as well as ideas for problem solving, dramatisation, modelling and direct experience of phenomena. Finally, the 'Creative Literacy' boxes contain motivational ways for students to record and express their knowledge.

This book is intended to encourage motivation in science, but it is not about big bangs and explosions nor does it emphasise the use of scientific apparatus. In fact, a conscious decision was made to include many activities that do not rely on specialised equipment. My belief is that if science is something that involves everyday materials (and activities that can be done at home in the kitchen) then this automatically makes it relevant to real life.

An important criterion for the selection of activities was that they can all be relied on to work with a minimum of fuss. However, like all science activities, they are not completely foolproof. Readers would be advised to abide by Dave's Golden Rule, which states 'Always, always, always, always, always, test it by yourself first, before you try to impress other people'. In this way you can check it with the materials you have available and adapt it for your conditions.

There are detailed safety instructions for any experiments that might involve risk factors. Please note, however, that this book is designed for the use of teachers in their classrooms, not for students in an unsupervised environment. Teachers should follow all normal safety procedures.

One issue that became apparent to me when I first began writing this book was that many of the activities relate to physical science concepts and that, by comparison, biology and chemistry are less well represented. However, I make no apology for this. I believe that different sciences are interesting for different reasons, and one of the features of physics is that it does lend itself to the creation of surprising hands-on activities. In fact, with recent reports that students' interest in physics is falling (especially among girls), it could be argued that it is in precisely this area that we need to develop interesting and motivational experiences.

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41 CARD AND CUP

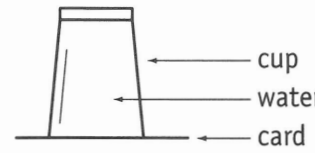
You can turn a cup upside down and the water won't come out.

HOW TO DO IT

1 Fill a glass with water and put a playing card over the top. The card should cover the top of the glass completely.

2 Put your finger on the card to hold it in place as you turn the cup upside down.

3 Remove your finger from the card. (*The card and water will not fall.*)



EXPLANATION

The external air pressure pushing on the card is greater than the downwards water pressure on the card.

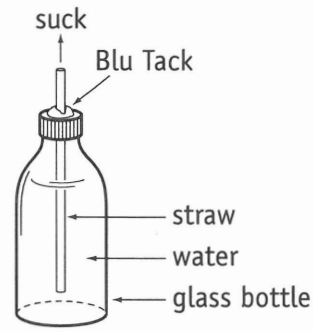
CREATIVE LITERACY Write a script for a 'science theatre' performance of the card and cup demonstration.

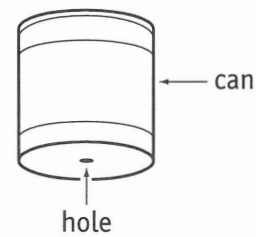
It is difficult to swat a housefly because it is extremely sensitive to changes in air pressure. As a rolled up newspaper descends towards one it can sense the wind of its movement in time to escape. Fly swatters have holes in them to reduce this effect.

LINKED ACTIVITIES

1 **Make a straw that doesn't work.** Make a hole in the lid of a glass bottle and insert a straw through it. Pack around the hole with Blu Tack to make it strongly airtight. Fill the bottle with water. See how much water you can drink with one big suck. *(The ambient air pressure is prevented from acting upon the water in the bottle, so you will initially suck up a small amount of fluid, and then no more.)*

2 **Can you pour water from one hole?** Put a small nail hole in the bottom of an empty paint can. Temporarily block the hole with a piece of tape before you fill the can with water and put the lid on tightly. Then remove the tape. *(The water will not pour out.)* Open the lid. *(The water will fall from the lower hole.)*





HISTORICAL NOTE

Canned food was available by about 1820, but the can opener was not invented until more than 40 years later. In the meantime, people used hammers and chisels to open their cans.

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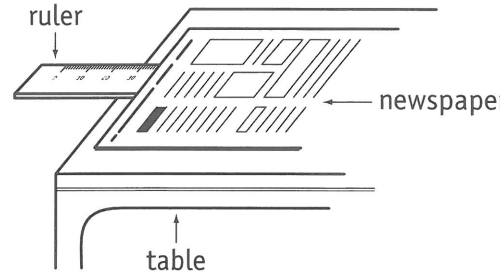
42 BREAK A RULER

A sheet of newspaper can break a ruler.

HOW TO DO IT

1. Place a cheap, light, wooden school ruler on a bench so that about one third of its length is projecting out from the edge of the bench.
2. Place a sheet of newspaper over the part of the ruler that is on the bench.
3. Hit the exposed end of the ruler very firmly. *(The ruler will break.)*



EXPLANATION

When you hit the ruler it begins to lift the newspaper. This momentarily reduces the air pressure under the newspaper, but the higher air pressure above the newspaper presses down and stops it lifting. Because the ruler is prevented from lifting, it breaks.

CREATIVE LITERACY Imagine that you were a person in the crowd who witnessed von Guericke's demonstration of air pressure (see Historical Note). Write a diary entry for that day.

HISTORICAL NOTE

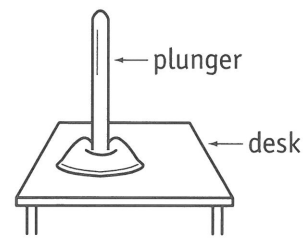
Air pressure is stronger than horse power. In 1654, Otto von Guericke put two metal bowls together to form a sphere about the size of a person's head. He then pumped the air out of the bowls so they were only held together by ambient air pressure. Sixteen horses were unable to pull the bowls apart.

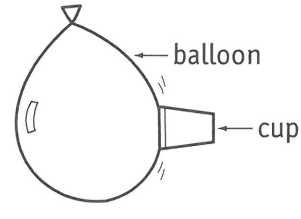
Tree frogs have suction cups on their feet for climbing.

Squid and octopus also have suckers. Giant squid often battle with sperm whales (their predators) in the ocean depths. By measuring the size of the sucker marks on the bodies of sperm whales, scientists have estimated that some giant squid are as long as a cricket pitch.

LINKED ACTIVITIES

1. **Lift a desk.** Obtain a large plumber's plunger, wet the rubber, and plunge it onto the top of a school desk. (This will be most effective if the desk has a smooth, laminated top.) Pull the plunger upwards. *(A partial vacuum will form between the plunger and the desk. The outside air pressure will be strong enough to keep the plunger attached to the desk as it is lifted.)*
2. **Put a nose on a balloon.** Hold the mouth of a plastic cup to the side of a balloon as you blow it up. When the balloon is blown up, let go of the cup. *(The balloon will seal the opening of the cup, and the shape of the seal will change from more rounded to less rounded as the balloon is inflated. This will increase the volume of the air inside the cup, thus reducing its pressure.)* Draw a face on the balloon.





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43 CRUSH A CAN

You can use invisible force to crush a can.

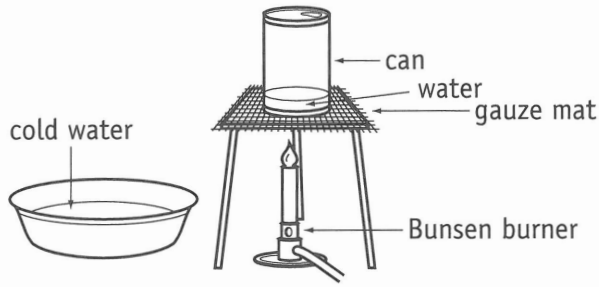
HOW TO DO IT

TEACHER DEMONSTRATION ONLY

Safety glasses and laboratory coat should be worn.

1 Boil a small amount of water in a used soft drink can.

2 Use a large pair of tongs to quickly lift the can towards a water tray and immediately dip the top of the can downwards into the cold water. (*The can will collapse with a loud bang.*)



EXPLANATION

The steam from the boiling water forces some of the air to leave the can.
When the can is suddenly cooled, the air inside it contracts, reducing the air pressure. The greater air pressure outside the can crushes it faster than the water can enter.

TO INVESTIGATE

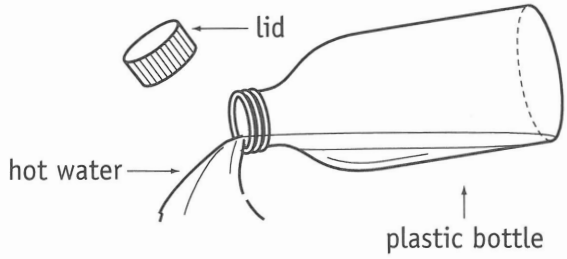
How much air pressure is acting on the outside of the can? Calculate the surface area of the can and find out the total pressure acting on it if air exerts about 1 kg of pressure per square centimetre. *(It is roughly equivalent to the weight of two or three people standing on the can.)*

HISTORICAL NOTE

Aristotle, the famous Greek philosopher, did not realise that wind is simply air that is moving.

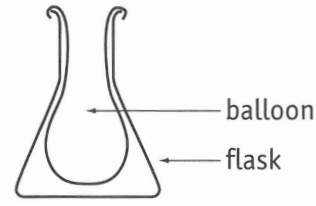
LINKED ACTIVITIES

1 **Crush a plastic bottle.** Pour some hot tap water into a plastic bottle. Swirl it inside the bottle then tip it out and immediately screw the lid on tightly. Run some cold tap water over the bottle. (*The bottle will be partially crushed.*)



CREATIVE LITERACY Why don't people notice air pressure? Write how you would convince a sceptic that air pressure does exist.

2 **Inflate a balloon inside a flask.** Rinse a 1L flask out with hot tap water then immediately fit a balloon over the mouth of the flask. *(As the air inside the flask cools and contracts, the balloon will be pushed inside out into the flask.)*



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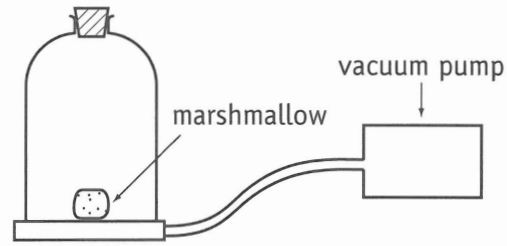
44 EXPAND A MARSHMALLOW

You can inflate a marshmallow.



HOW TO DO IT

Safety glasses should be worn.

1. Place a marshmallow inside a bell jar.
2. Seal the bell jar and connect it to a vacuum pump.
3. Pump the air out of the bell jar. *(The marshmallow will inflate.)*

EXPLANATION

As the bell jar is evacuated, the air in pockets inside the marshmallow have a higher pressure than the air in the bell jar. The marshmallow gets bigger as the air inside it expands.

TO INVESTIGATE

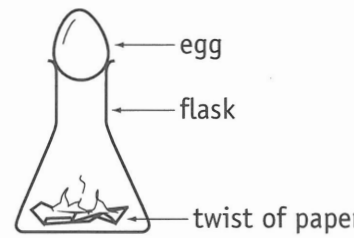
What would happen if shaving cream was substituted for the marshmallow?

Up in space there is no air, so there is no air pressure. If people went up into space without a pressurised suit, their blood would boil and their body expand like a marshmallow.

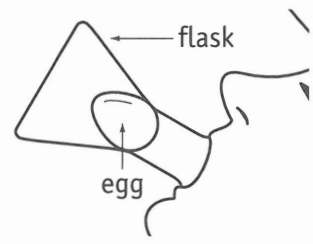
Outer space is not a total vacuum. Each litre of outer space contains about 100 molecules of gas.

LINKED ACTIVITIES

1. **Make an egg go into a bottle.** Select an egg that is only slightly larger than the mouth of a 1L conical flask. Hard boil the egg, then shell it and keep it wet. Drop a burning twist of paper into the 1L conical flask. Place the wet egg in the mouth of the flask. *(The burning paper heats the air in the flask, so some of the air is forced out by expansion. As the air cools it contracts, and the greater ambient air pressure will force the egg into the flask.)*



2. **Make an egg go out of a bottle.** To recover the egg in the flask above, firstly turn the flask upside down so the egg is fitting snugly into the neck. Then put your lips around the mouth of the flask, to seal it tightly, and blow as hard as you can into it. *(When you stop blowing, the egg will move out of the flask and into your mouth, as you have increased the pressure inside the flask.)*



CREATIVE LITERACY Videotape a person performing the two activities above. Write a script for a voice-over that could be recorded onto the video.

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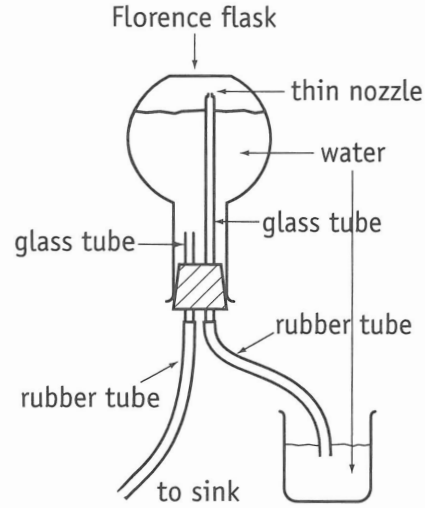
45 SIPHON FOUNTAIN

Air pressure can make water flow upwards.



HOW TO DO IT

1. Add water to a Florence flask until the water level reaches the neck of the flask.
2. Obtain a stopper with a long glass tube (with a narrow mouth) and a short glass tube already inserted. (*Safety warning: the glass tubes should be inserted into the stopper by an experienced laboratory assistant because there is a risk of the glass breaking*). Attach rubber tubing to each one, as shown in the diagram.
3. Insert the stopper into the mouth of the flask, then invert it.

EXPLANATION

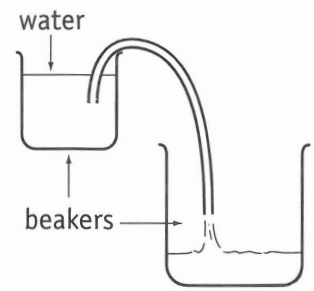
As the water in the flask drains into the sink, a partial vacuum is created in the flask. Ambient air pressure pushes the water from the beaker into the flask.

Inside the funnel of a tornado the air pressure is reduced by about 90%. When tornados occur over water they result in waterspouts. Sometimes, fish and frogs have been sucked up in waterspouts, carried along, and dropped down on neighbouring farms.

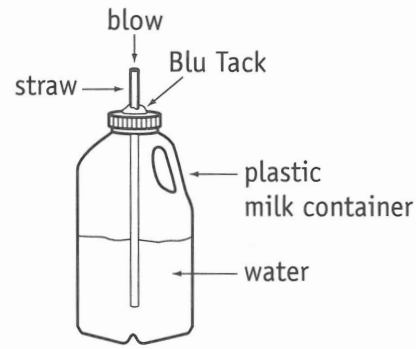
CREATIVE LITERACY Write an explanation in 'kidspeak' for the siphon fountain activity.

LINKED ACTIVITIES

1. **Make a siphon.** Hold a 200 mL beaker full of water on top of the bench so it cannot tip over. Position an empty, larger beaker below the bench. Place one end of a rubber tube in the water, suck the other end and immediately place it in the lower beaker. *(The water will drain from the upper beaker to the lower because the sucking creates a partial vacuum in the tube, and the ambient air pressure pushes the water into the tube.)*



2. **Make a fountain with a milk container.** Take a 2 L plastic milk container, put a small hole in the lid, push a strong plastic straw into the hole and make the join airtight by lining it with Blu Tack. Half fill the container with water, and put the lid on tightly. Blow strongly through the straw into the container. *(The greater air pressure inside the bottle will force the water to spray out of the straw.)*



CREATIVE LITERACY How do water pistols work? Buy a cheap water pistol and carefully examine it. Pull it apart if necessary. Prepare a series of labelled diagrams to show how it works.

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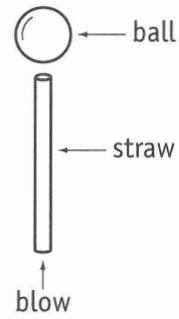
46 BLOW A BALL

You can keep a ping-pong ball suspended in the air.



HOW TO DO IT

1. Blow through a straw which you are holding vertically upward.
2. Put a ping-pong ball in the stream of moving air above the straw. Blow strongly and keep the straw vertical. (*The ping-pong ball will remain suspended just above the straw.*)

EXPLANATION

A fast moving stream of air has lower pressure, so the pressure above the straw while you are blowing is lower than in the normal air around it. Air moves from high pressure to low pressure, so the ball is constantly pushed into the area directly above the straw where the pressure is lower.

TO INVESTIGATE

Place the ping-pong ball on the bench and use the straw to blow a stream of air to one side of the ball. *(The ball will initially move towards the stream of air because the pressure is lower there.)*

When you are standing on the kerb by the side of the road and a large truck passes by at speed, you feel as if you are sucked towards the road. This is partly due to the Bernoulli effect created by the moving streams of air.

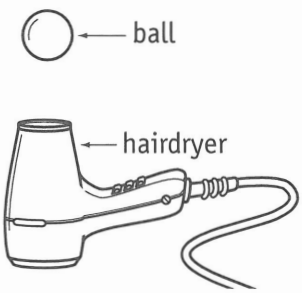
The tarpaulin over the back of a truck is flat when the truck is not moving, but it bulges upward when the truck is moving because of the air flowing over the top of it.

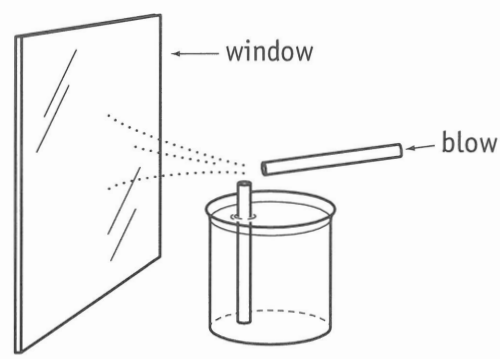
HISTORICAL NOTE

Daniel Bernoulli (1700–1782) was initially taught mathematics by his father, but he got so good at it that his father became jealous and is said to have thrown him out of the house.

LINKED ACTIVITIES

1. **Use a powerful hairdryer to suspend a ping-pong ball in the air** (use the cold button).
2. **Make a spray gun.** Hold a straw in a beaker of water so it projects about 2 cm out of the water. Use another straw to blow at the top of the first straw. *(If you do this activity in front of a window then a fine spray of water will hit the glass. The fast moving stream of air creates lower pressure above the vertical straw. The higher ambient air pressure acts on the water in the beaker, forcing some of it up the straw.)*





CREATIVE LITERACY Write an advertising blurb extolling the features of your straw spray gun.

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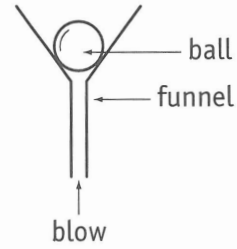
47 BALL IN A FUNNEL

You cannot blow a ping-pong ball out of a filter funnel.



HOW TO DO IT

1. Put a ping-pong ball into a plastic filter funnel.
2. Hold the funnel vertically upright and blow upwards through it. (*The ball will remain in the funnel.*)

EXPLANATION

The air you are blowing is flowing between the ball and the sides of the funnel. This fast moving air has lower pressure than the air outside the funnel, so the ball is constantly pushed into this area where the pressure is lower.

TO INVESTIGATE

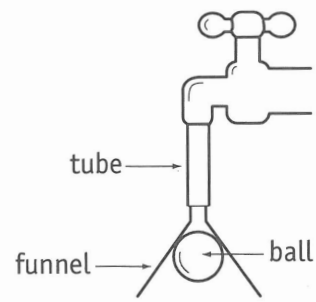
Predict what would happen if you turned the funnel upside down while blowing through it. (*The ball will not fall out of the funnel as long as you blow strongly.*)

If a person is smoking inside a moving car, a good way to get rid of the smoke is to open the window slightly. There is a stream of air flowing down the outside of the car, so the air pressure is relatively lower outside and the smoke is forced out of the window.

The Bernoulli effect also operates in water. If two ships are travelling in the same direction and close to each other, there will be a fast stream of water flowing between them. The water pressure will therefore be lower between the ships and they will tend to move towards each other, risking a collision.

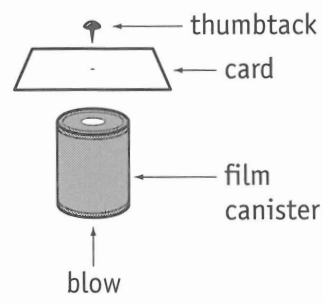
LINKED ACTIVITIES

1. **Demonstrate the Bernoulli effect in water.** Put a ping-pong ball in a plastic filter funnel and use a rubber tube to connect the funnel to a tap. Hold the ball in place as you turn the water on, then let it go. (*The ball will remain in the funnel.*)



CREATIVE LITERACY Imagine that you are the ping-pong ball in the activity above. Write an account of the forces acting upon you.

2. **Blow a card.** Make a pencil-sized hole in the base of a film canister. Insert a thumbtack through the centre of a playing card, then loosely position its point in the hole in the canister. Blow through the canister from the other end. (*There is a low pressure area between the card and the canister because the air is moving quickly there. The higher air pressure on the other side of the card will hold it in place.*)



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48 FLYING COIN

You can make a 5c coin flip into the air by blowing it.

HOW TO DO IT

1 Place a 5c coin and a dish on a table. The coin should be about 5 cm from the edge of the dish.

2 Get down at table height and direct a quick, strong blow across the top of the coin. *(With a bit of practice it will flip up and land in the dish.)*

EXPLANATION

The moving air produces lower pressure above the coin. This lifts the coin in much the same way as an aircraft wing is lifted by the air flowing over the top of it.

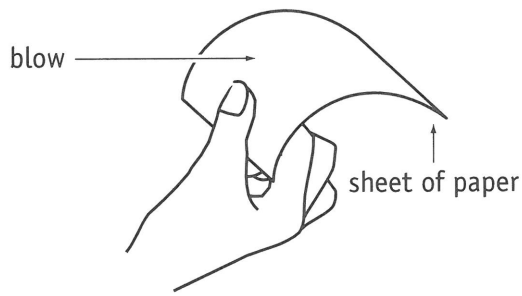
Hummingbirds can fly backwards.

HISTORICAL NOTE

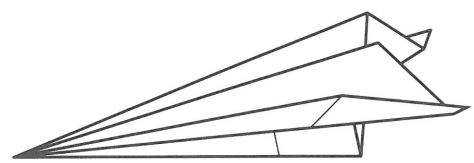
In earlier days when aircraft were less powerful, they would sometimes become stationary in the sky if they were flying into a strong headwind.

LINKED ACTIVITIES

1 **Demonstrate how aircraft wings achieve lift.** Hold a sheet of paper as shown in the diagram, and blow across the top of it. *(When air flows over the top of the paper, it will rise because the moving air is a region of reduced pressure. The top of an aircraft wing is curved like the sheet of paper.)*



2 **Solve the following problem.** Design a paper aeroplane that will turn around and come back towards you. *(Only about 60% of the lift of an aeroplane is caused by the Bernoulli effect. The rest is caused by the slope of the wings, as in a kite or paper aeroplane.)*



CREATIVE LITERACY Write a set of instructions for making the paper aeroplane you designed. Test the quality of your instructions by asking another student to follow them to make an exact replica of your paper aeroplane.

Acrobatic aircraft do not use the Bernoulli effect at all because they sometimes have to fly upside down. They have symmetrical wings and always have to fly nose up to get lift.

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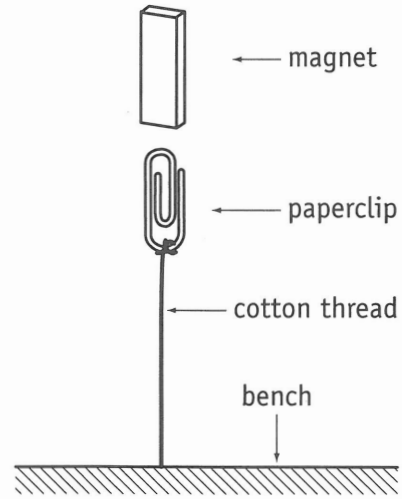
49 FLYING PAPERCLIP

A paperclip can be suspended in midair.



HOW TO DO IT

1. Cut about 30 cm of light cotton thread. Attach one end to the bench and tie the other end to a paperclip.
2. Lift the paperclip up with a magnet until the cotton is taut and then move the magnet slightly further away. (*The paperclip will be suspended in midair.*)
3. Move a sheet of paper through the space between the magnet and the paperclip. (*The paperclip will remain suspended.*)

EXPLANATION

The steel paperclip does not have to touch the magnet in order to be attracted to it — it just has to be within the magnetic field. The magnetic field can pass through nonmagnetic material such as paper.

TO INVESTIGATE

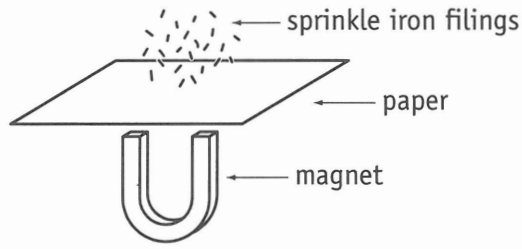
Which other types of materials will the magnetic field pass through? Predict whether the paperclip will remain suspended if wood, plastic, aluminium foil, glass or steel are placed between it and the magnet.

Some people have enough iron in their bodies to set off airport metal detectors. These people have a condition called haemochromatosis, a genetic disorder that results in iron overload.

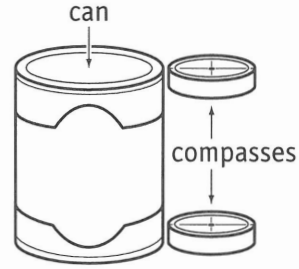
Many migratory animals such as pigeons and salmon have tiny magnetic particles in their brains, and they appear to use the earth's magnetic field for navigation. Strangely, we humans have these particles too, but scientists don't know why.

LINKED ACTIVITIES

1. **Make magnetic artwork.** Position a magnet under a sheet of light cardboard, then sprinkle iron filings onto the cardboard and tap it lightly. *(The iron filings will form a pattern representing the lines of force of the magnetic field.)* Spray it with clear varnish to preserve the pattern.



2. **Investigate the magnetism of food cans.** Look in the back of your kitchen cupboard for a food can that has been there for several weeks. Put it on a table, then hold one compass next to the top of the can and another next to the bottom. *(The compass needles will point in opposite directions because the can has been magnetised by the earth's magnetic field.)*



Every half a million years or so the earth's magnetic poles swap over.

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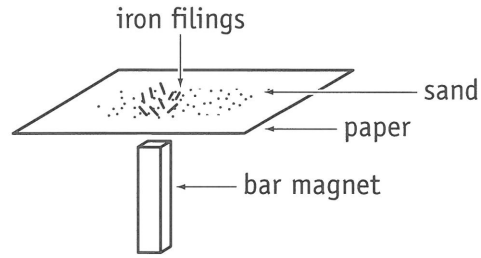
50 MAGNETIC ECHIDNA

You can make iron filings stand up and move around on a sheet of paper.

HOW TO DO IT

1 Place a teaspoonful of sand/iron filings mixture on a sheet of paper.

2 Move a bar magnet under the paper to separate the mixture into two heaps. (*The iron filings will stand up and follow the magnet, leaving the sand behind.*)



EXPLANATION

The magnet attracts iron filings but not sand. The magnetic field can pass through the paper.

CREATIVE LITERACY Design a board game in which objects on top of a sheet of cardboard are manipulated by magnets underneath it. Write a set of instructions for the game.

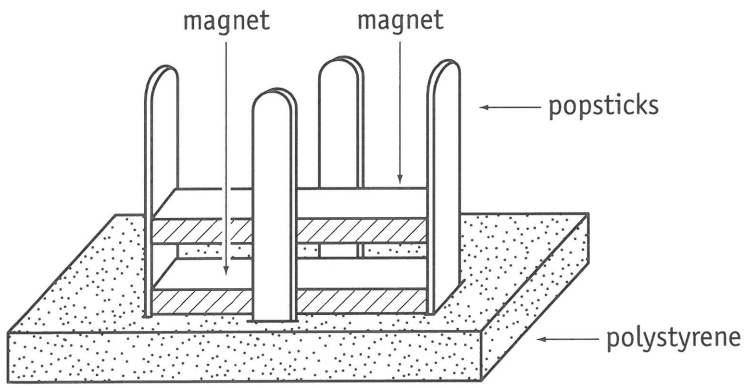
TO INVESTIGATE

What thickness of paper or other material can the magnet operate through?

The north pole of a bar magnet is actually a south pole, because it will point to the earth's north magnetic pole if it is allowed to swing freely.

LINKED ACTIVITIES

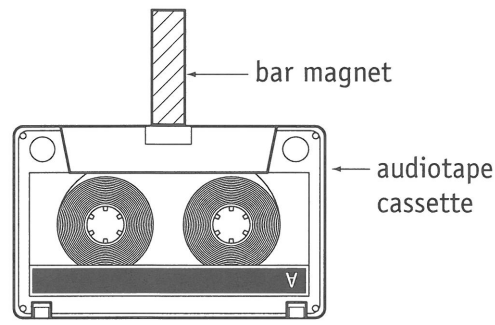
1 **Levitate a magnet.** Place a bar magnet on a thick slab of polystyrene foam. Insert a popstick into the foam at each end of the magnet, and two more on each side. Slip another magnet, similarly aligned, into the corral formed by the popsticks. (*Repulsion will cause it to float above the other magnet.*)



HISTORICAL NOTE

Christopher Columbus's compass malfunctioned when his ships entered the area of the Bermuda Triangle. He also observed strange lights in the sky in this area.

2 **Erase an audiotape.** Record the sound of your voice on an old audiotape. Eject the audiotape from the machine and run a strong magnet along that section of the tape. (*An audiotape has magnetic particles on its surface, and these are pulled into patterns when sound is recorded. However, a magnet will erase the patterns.*)



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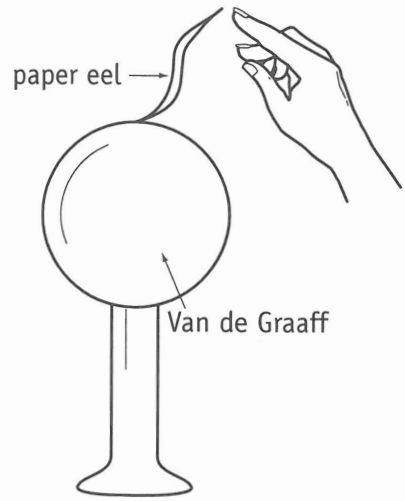
51 ELECTRIC EEL

You can make a paper eel strike at your hand.



HOW TO DO IT

TEACHER DEMONSTRATION ONLY

1. Cut an eel shape about 30 cm long and 1 cm wide from thin writing paper.
2. Use sticky tape to attach its tail to the dome of a Van de Graaff generator.
3. Switch the generator on low speed only. *(The eel will rise up into the air.)*
4. Extend your hand towards the electric eel. *(It will strike at your hand.)*

EXPLANATION

The paper eel becomes charged; as charged objects are attracted to uncharged objects, it moves towards your hand.

A real electric eel can produce enough electricity to power all the light bulbs in a four bedroom home. These eels live in South America and can be over 2.5 m long.

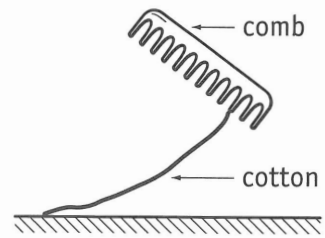
When platypuses swim underwater, they close their eyes, ears and nostrils, yet they can still find food. They do this by detecting minute electrical discharges given off by organisms in the water.

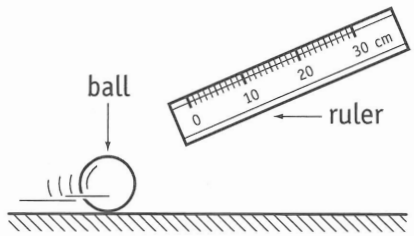
HISTORICAL NOTE

During the 1960's, people who suffered from manic depression were sometimes treated with electric shocks. These shocks were severe enough to cause convulsions.

LINKED ACTIVITIES

1. **Make a snake on your desk.** Cut about 40 cm of cotton thread and tape one end to the desk. Bring a rubbed comb close to the cotton. *(The comb will attract the cotton.)*
2. **Attract a ball.** Rub a plastic ruler with wool. Bring the rubbed part of the ruler close to a ping-pong ball that is on the desk. *(Electrostatic attraction will make the ball roll towards the ruler.)*





CREATIVE LITERACY Imagine that you are an electron on the wool (see left). Describe your experiences throughout this activity.

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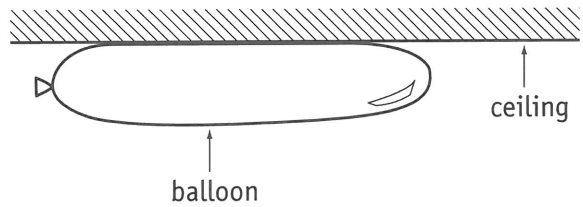
52 CHARGED BALLOONS

A balloon that has been rubbed will stick to the ceiling.

HOW TO DO IT

1. Blow up a sausage-shaped balloon and rub one side of it vigorously with a sheet of coarse paper towelling.
2. Hold the balloon so the rubbed part is touching the ceiling. Let it go. (*The balloon will stay there.*)



EXPLANATION

The balloon has become negatively charged by rubbing. Charged objects are attracted to uncharged objects, so the balloon is attracted to the ceiling. This force of attraction is strong enough to prevent the balloon falling to the floor.

TO INVESTIGATE

Will a rubbed balloon also stick to other materials, such as glass, curtains, timber, bricks, metal and people?

A typical lightning strike goes downwards AND then upwards, but it happens so quickly that our eyes are unable to separate the two movements.

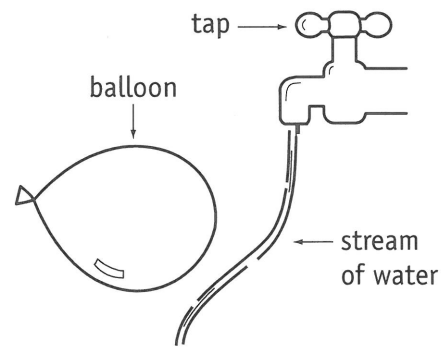
Roy Sullivan, a park ranger in Virginia USA, has been struck by lightning seven times and is still alive.

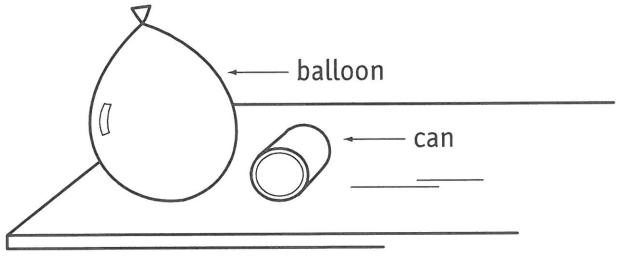
HISTORICAL NOTE

Benjamin Franklin investigated electricity by flying a kite in a thunderstorm. However, scientific experiments are meant to be reproducible, and unfortunately the German scientist who tried to reproduce this experiment was electrocuted and died.

LINKED ACTIVITIES

1. **Use a balloon to bend tap water.** Turn a tap on to produce a steady, unbroken thin stream of water. Bring a rubbed balloon close to the stream of water. *(The stream of water will bend as it is attracted towards the charged balloon.)*
2. **Use a balloon to attract a can.** Rub a balloon with coarse paper towelling and bring it close to an empty aluminium can that is lying on its side. *(The can will follow the balloon across the bench.)*





CREATIVE LITERACY Describe a real life situation in which you experienced static electricity. Write a scientific explanation for why it happened.

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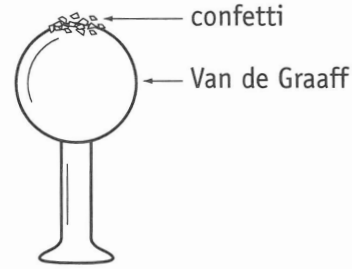
53 FLYING CONFETTI

Static electricity can make confetti take off into the air.



HOW TO DO IT

1. Place a handful of confetti on top of the dome of a Van de Graaff generator.
2. Switch the generator on. *(The confetti will rise into the air and scatter in all directions.)*

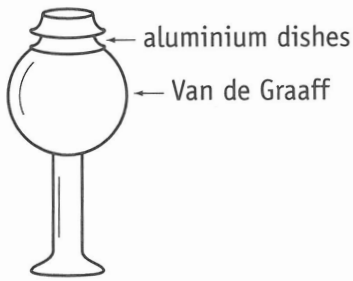
EXPLANATION

Like charges repel. The pieces of confetti become charged with the same charge as the generator, so they are repelled by the generator and by each other.

Static electricity experiments work best on dry days. On damp days, some of the charge tends to leak into the humidity in the air.

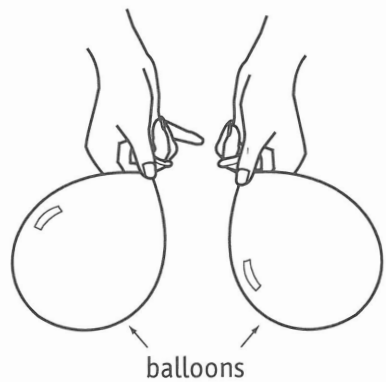
LINKED ACTIVITIES

1. **Repel aluminium dishes.** Place an aluminium dish upside down on the dome of the Van de Graaff generator and tape it in position. Place another dish on top of it, then turn the generator on. *(The second dish will rise up and slip off the dome because it picks up the same charge as the first.)*



CREATIVE LITERACY Place a nylon wig on top of the Van de Graaff generator. Write a scientific explanation for what happens.

2. **Repel balloons.** Blow up two balloons and rub both of them vigorously with paper towelling. Hold the balloons by the nozzles, so they can hang and swing freely. Bring them near to each other. *(They will repel each other because they have the same charge.)*



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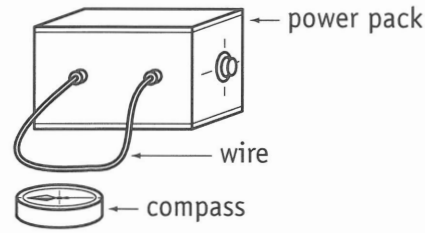
54 ELECTROMAGNETS

An electric current can stop a compass from pointing north.

HOW TO DO IT

1. Use an insulated wire to connect the two terminals of a power pack, with the switch off.
2. Place a pocket compass next to the wire so that the wire and the compass needle are parallel.
3. Set the power pack to 2 V and briefly switch on the current. *(The compass needle will swing around to point away from the wire.)*



EXPLANATION

A current flowing along a wire creates a magnetic field around it. The direction of the current determines the direction of the magnetic field.

TO INVESTIGATE

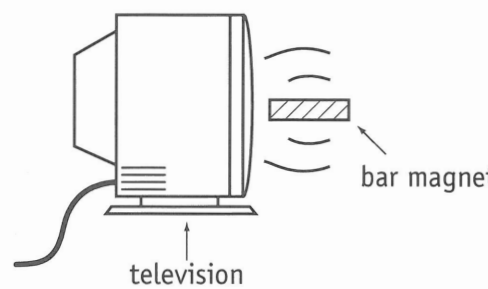
What would happen if the direction of the current was reversed by switching the terminals on the power pack? What would happen if the compass was moved to the other side of the wire?

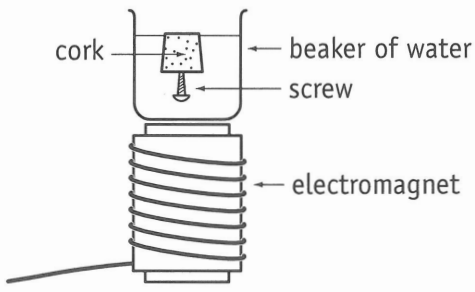
The sun is an electromagnet because of the movement of charged particles in its interior. Interestingly, its poles swap every 11 years.

The earth's magnetic poles are moving at a rate of about 10 km each year. The southern magnetic pole is currently located on the coast of Antarctica, and the northern magnetic pole is located in northern Canada.

LINKED ACTIVITIES

1. **Distort a TV show.** *Warning: try this on an old TV.* Switch on your TV and slowly move a strong bar magnet across the front of the screen. *(The TV image is caused by electrons bombarding the screen. Moving electrons create a magnetic field, so the picture will become distorted when a magnet is brought near it.)*
2. **Sink a cork.** Insert a large steel screw into one end of a cork, then cut away some of the cork so it only just floats. Put it in a small beaker of water and place it securely on top of a commercial electromagnet. Switch the power on. *(The cork will sink as the electromagnet attracts the screw.)*





CREATIVE LITERACY Write a newspaper article that could have reported the invention of the electromagnet.

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55 ELECTRICITY FROM A KIWI FRUIT

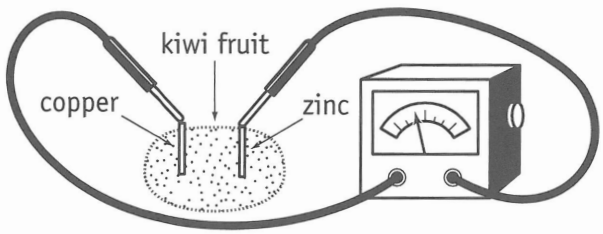
You can make electricity with fruit.

HOW TO DO IT

1 Use wires to connect a copper plate to one terminal of a microammeter and a zinc plate to the other.

2 Cut two slits, about 2 cm apart, in a kiwi fruit.

3 Insert the two metal plates into the slits. *(The microammeter will indicate a flow of current.)*



HISTORICAL NOTE

One of the scientists who helped to develop batteries was Robert Bunsen. While doing research in chemistry, he nearly died from arsenic poisoning and lost the sight of one eye in a laboratory explosion. The Bunsen burner was named after him although he didn't initially design it.

EXPLANATION

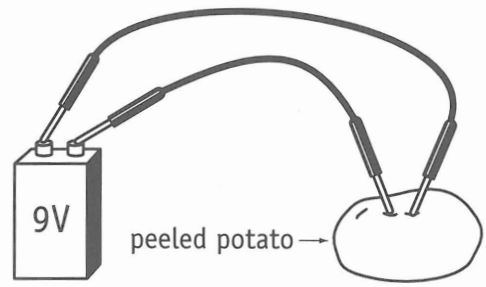
The zinc attracts electrons from the citric acid in the kiwi fruit so it becomes negatively charged. The copper donates electrons to the solution and becomes positively charged. The difference in charge causes electrons to flow through the wire from the zinc to the copper.

TO INVESTIGATE

Do some fruits generate more current than others? To what extent is the amount of current influenced by the depth of the electrode in the fruit?

LINKED ACTIVITIES

1 **Use a potato to test for a flat battery.** Attach two wires to the terminals of a 9 V household battery. Insert the other ends of the wires into a peeled potato so they are about 2–3 mm apart. *(If the battery is charged, bubbles of hydrogen gas will appear around the wire connected to the negative terminal; after a minute or so a blue colour will appear at the wire connected to the positive terminal.)*



2 **Solve the following problem.** Is it possible to recharge a flat battery by leaving it in the sun for a while? Design an experiment to test this question.

3 **Solve the following problem.** Battery manufacturers sometimes claim that their products last longer than those of other brands. Test their claims by inserting batteries into electrical toys (or other devices with moving parts such as battery shavers) and comparing the time taken to run down. Calculate the brand that offers the best value for money.

CREATIVE LITERACY

Choice magazine regularly reports on products available to consumers. Write a report about the batteries you tested in the activity (left) and send it to the magazine.

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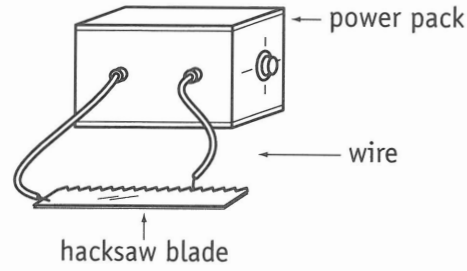
56 ELECTRIC SAW

An electric current can make sprays of sparks.

HOW TO DO IT

1. Attach two insulated wires to the terminals of a power pack set on 4 V.
2. Attach the bare end of one wire to the end of a hacksaw blade.
3. Darken the room and slowly run the bare end of the other wire along the hacksaw blade. *(Small sprays of sparks will be produced.)*



EXPLANATION

The intermittent flow of current produces a series of small electric sparks, which strike the steel, causing some of the steel particles to become white hot and break free from the blade as a visible spray of sparks.

Electrons in a wire only move a few centimetres each second. The effect however is instantaneous because all the electrons move at the same time.

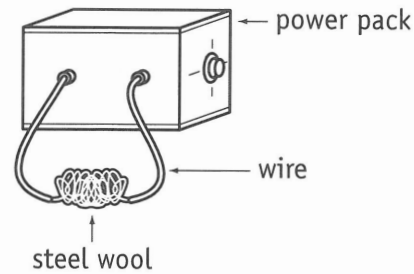
CREATIVE LITERACY Write a humorous story about a day in the life of a family whose members are not aware of safety with electricity. Include as many breaches of electrical safety as you can.

HISTORICAL NOTE

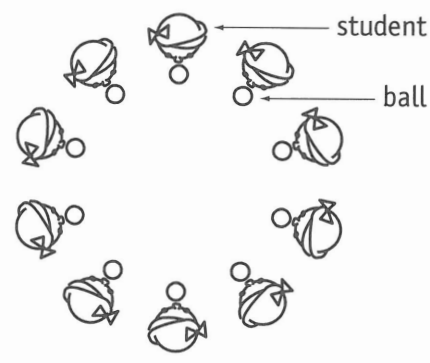
In the eighteenth century, electricity was such a novelty that some entertainers were able to make a living by demonstrating its effects. As a result a lot of innocent birds and animals and unsuspecting members of the public received nasty electric shocks.

LINKED ACTIVITIES

1. **Electrify steel wool.** *TEACHER DEMONSTRATION ONLY* Attach a power pack to a 3 cm twist of steel wool. Set the power pack to 10 V and switch it on. *(The steel wool will glow red hot and begin to disintegrate from the heat produced by the current.)*



2. **Dramatise the movement of electrons around a circuit.** Make a circle of 10–12 students, each of whom has a screwed up ball of newspaper. One nominated student should represent the battery, the other students represent the atoms in the wire, and the paper balls represent the free electrons. Dramatise a flow of current, then increase the 'voltage' or add a 'resistor'.



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57 PRESSURE SWITCH

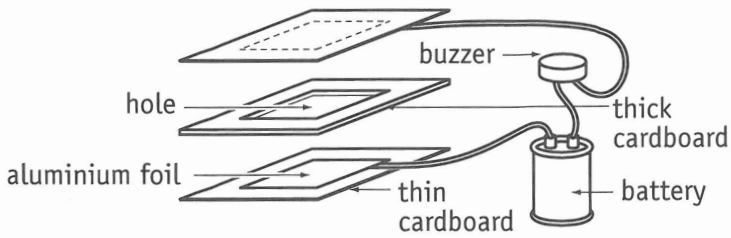
Some switches don't look like switches.

HOW TO DO IT

1 Tape aluminium foil to one side of a piece of thin cardboard. Tape the exposed end of a wire to the aluminium. Repeat this process to make another one. Cut a hole in a thick piece of cardboard and place it between the other two. This completes the pressure switch.

2 Use wires to connect the pressure switch to a buzzer and a battery.

3 Place the pressure switch flat on the bench and press the top piece of cardboard. *(The buzzer will sound.)*



CREATIVE LITERACY Design a burglar alarm circuit (i.e., a switch that will set off a buzzer when a door is opened). Write an advertising blurb explaining its excellent features.

EXPLANATION

The thick, central piece of cardboard separates the two sheets of aluminium foil. The circuit is not complete until the two sheets of foil are pressed together.

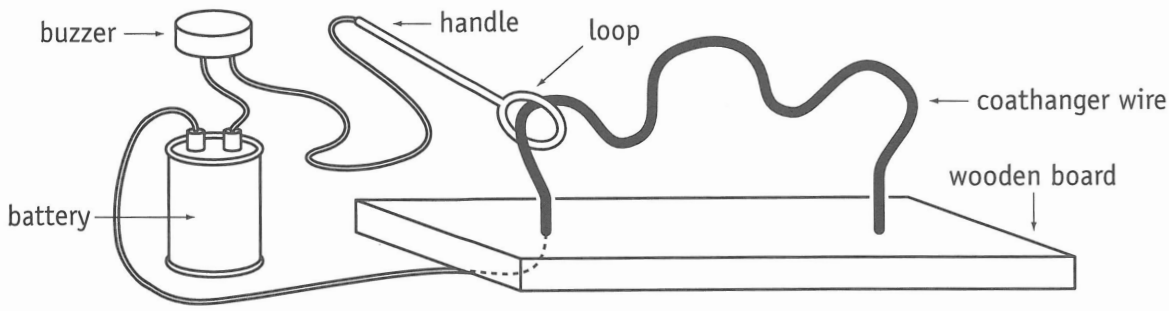
The wiring in satellites is often made of gold. This is because gold is a better conductor than copper.

HISTORICAL NOTE

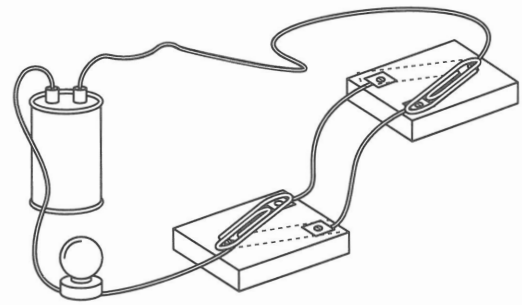
Thomas Edison left school at the age of 12. However, he still managed to invent the electric light bulb, the microphone, the generator and over 1000 other devices before he died in 1931.

LINKED ACTIVITIES

1 **Make a steady hand tester**, as shown in the diagram. *(The buzzer will sound when the loop touches the bare wire and completes the circuit.)*



2 **Build the switch** which is shown in the diagram and test it. Propose a use for it. *(Many modern houses have two switches to control one light, as this one does.)*



One of the quirks of science is that electrons flow from negative to positive, but current is taken as flowing from positive to negative.

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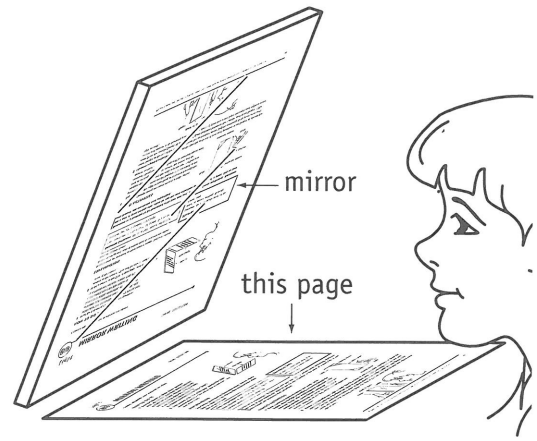
58 MIRROR WRITING

A mirror will jumble some words but not others.

HOW TO DO IT

1 Read the following multiple choice question:
WHY DID THE MUGGER RUN AWAY FROM ME?
a) SOME BYSTANDERS BEGAN HELPING ME.
b) POLICE WERE ARRIVING AT THE SCENE.
c) HE DECIDED I KICKBOXED OK.
d) A SIREN SOUNDED DOWN THE STREET.

2 Hold a mirror at the TOP of this page, and find the right answer in the mirror. *(You will be able to read item c, but none of the others.)*



EXPLANATION

Some letters are reversed in their mirror image, but not others. Item c is composed only of letters that do not reverse when they are upside down.

CREATIVE LITERACY Write a fact about reflection, then translate it into a form of code. Try to crack another person's coded message.

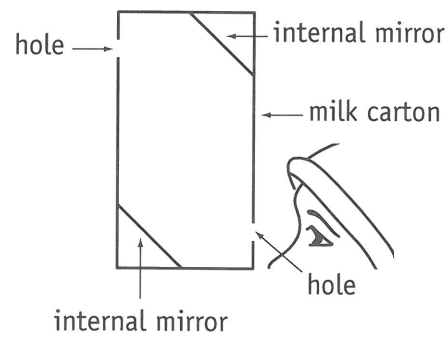
HISTORICAL NOTE

Leonardo da Vinci made most of his notes in mirror writing. Nobody knows why.

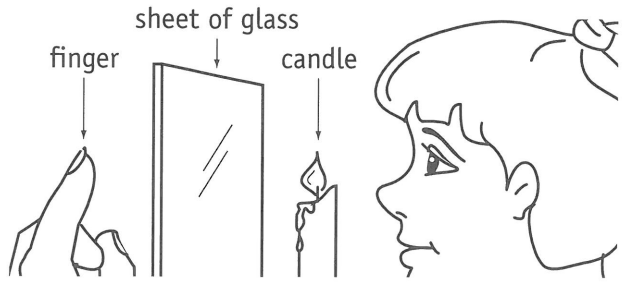
Narcissism is a tendency to be fascinated with your own self. In ancient Greek mythology, Narcissus was a man of great beauty who, when he saw his own reflection in the water of a pond, fell in love with himself and eventually died unfulfilled.

LINKED ACTIVITIES

1 **Make a periscope** using two mirrors and a milk carton. Explain where the light is travelling as it moves through the periscope. Predict what would happen if you looked at the mugger question (above) through a periscope. *(When you look at writing through one mirror, the letters will be reversed. Using two mirrors, they will be reversed again and appear normal.)*



2 **Observe a partial reflection.** Light a candle directly in front of you and position your finger at about the same height as the tip of the flame but about 30 cm further away (*Safety warning: keep your arm well away from the candle flame*). Hold a sheet of glass halfway between the candle and your finger. *(The candle will appear to be burning your finger. The sheet of glass lets some light through and reflects some light as well.)*



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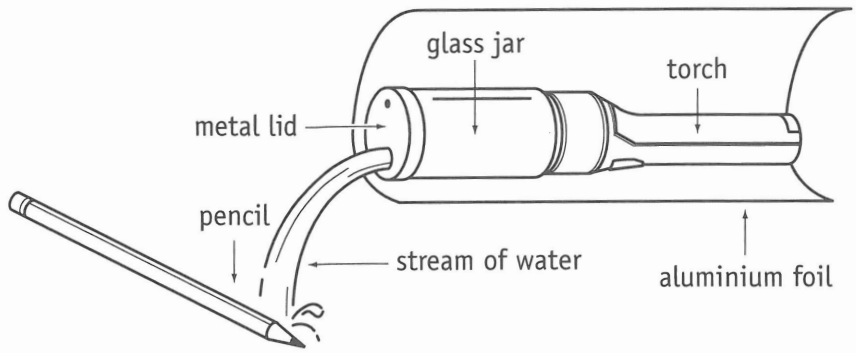
59 STREAM OF LIGHT

You can trap light inside a stream of water.

HOW TO DO IT

1. Obtain a glass jar with a screw-on metal lid. Make a small hole near one side of the lid and a larger one (0.5 cm diameter) towards the opposite side. Fill the jar with water and put the lid on.
2. Switch on a torch and position it so it shines through the base of the jar.
3. Cover the torch and the jar with a large piece of aluminium foil so that no light is escaping. Darken the room.
4. Tip the jar so a steady stream of water is coming from the larger hole.
5. Place a pencil in the stream of water at some distance from the jar. *(There will be a spot of light on the pencil as long as it is in the stream of water.)*

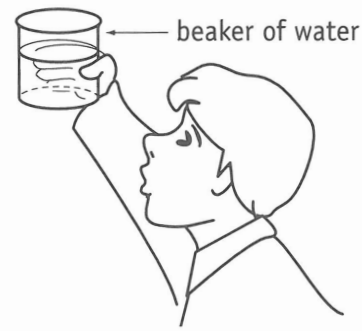


EXPLANATION

Light rays travelling through water are reflected at the surface of the water, if they have approached it at an acute angle. The surface of the water acts like a mirror, so the light keeps bouncing from one side to another inside the stream of water.

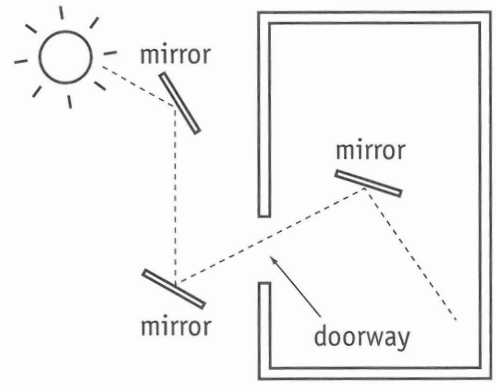
A standard optical cable is capable of transmitting the entire contents of ten large phone books in less than a second.

CREATIVE LITERACY Write a rhyming ditty that ends with the words 'total internal reflection'.



LINKED ACTIVITIES

1. **Experience total internal reflection.** Hold a beaker of water above your head and try to look through the surface of the water from underneath. *(When you look at the surface from an acute angle, you will not be able to see through it because it acts as a mirror.)*
2. **Model total internal reflection.** Hold a mirror in a direct beam of sunlight near a window. Direct the light to a second mirror in another part of the room. Use other mirrors to transfer the light around a corner and into another room.



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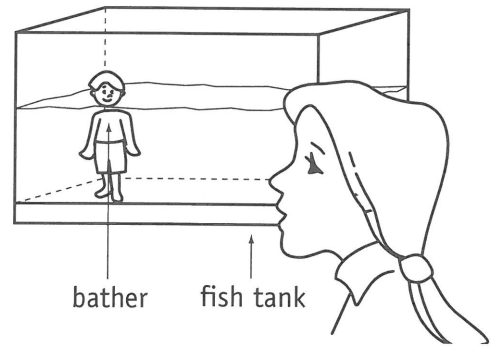
60 FISH TANK BATHER

A figure in a fish tank will appear to be cut in half.

HOW TO DO IT

1. To make the bather, cut out the shape of a person about 15 cm high from stiff cardboard.
2. Lower the bather up to its neck in the water in a fish tank, near to the front glass.
3. Sit at an acute angle to the front of the tank, and observe the bather. *(Its head will appear to be disconnected from its body.)*



EXPLANATION

The light image coming from the submerged part of the bather is being refracted by the water, but the image from the bather's head is not. To an observer who is at an angle to the glass, the image of the body seems to be coming from a different place.

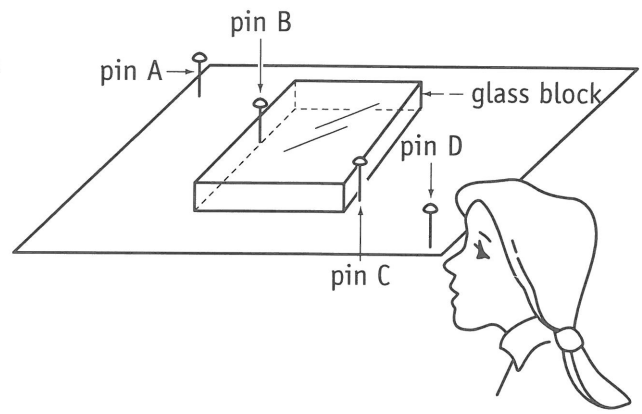
Light slows down to just over half its normal speed when it goes through glass.

You can see the sun just after it has set. The light from the sun becomes bent downwards as it enters the earth's atmosphere. This means that the sun can appear to be higher in the sky than it actually is. So we can still see the image of the sun after it has actually sunk below the horizon.

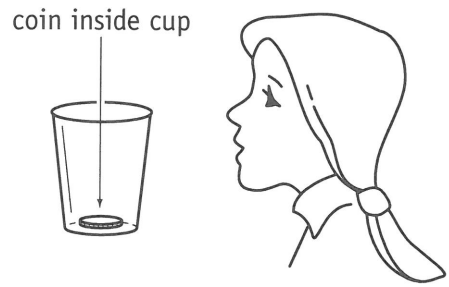
CREATIVE LITERACY Why is it difficult to spear a fish from above the water? Write a set of instructions that would help a person to do this more accurately.

LINKED ACTIVITIES

1. **Track the path of an image.** Place a glass block on a new page of a workbook. Insert a pin A about 5 cm from the block, as shown. Observe pin A through the glass from the other side, and while doing so insert pins B, C and D so they all appear to line up. Trace the outline of the glass slab, then remove it, and draw straight lines to connect the pins. *(The four pins will not be in a straight line. You can see that the light was bent as it moved from pin B to pin C.)*



2. **Use water to make a coin appear.** Place a dark coin in the bottom of a polystyrene cup. Stand to one side so that the coin is just out of your view. Slowly add water to the cup, but keep your head in the same position. *(The coin will reappear as the cup fills up. Light rays from the coin are refracted as they leave the water, so the image of the coin is coming from a different angle.)*



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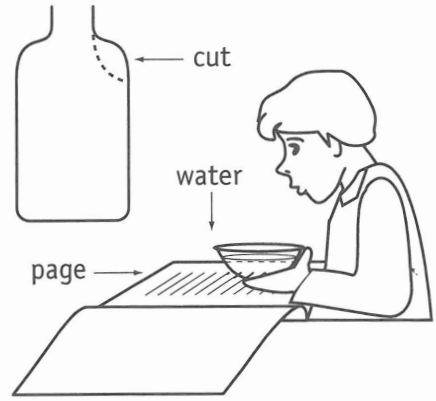
61 MAGNIFYING

Water can magnify objects.



HOW TO DO IT

1. Cut a circular piece of clear plastic about 5–6 cm in diameter from the shoulder of a clear plastic bottle. It should be shaped like a hemisphere.
2. Hold the piece of plastic like a cup and fill it with water.
3. Hold it above this book and look down through the water at the print. *(The print will be magnified.)*

EXPLANATION

The body of water has the shape of a convex lens. Light from the print becomes refracted as it enters the water, so the observer sees an enlarged virtual image.

TO INVESTIGATE

What would happen if you used a less rounded piece of plastic?

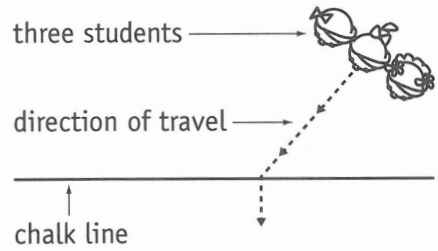
A telescope is a device for seeing back into time. The light we see from distant stars was produced millions of years ago.

HISTORICAL NOTE

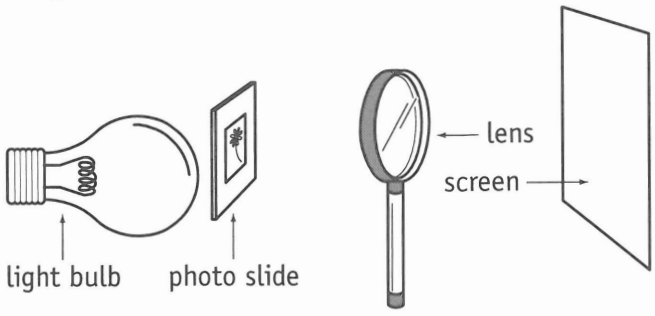
Antonie van Leeuenhoek (1632–1723) was a draper who, as a hobby, made high quality lenses for microscopes. He was the first person to observe bacteria and human sperm. However, like many of his contemporaries, he believed that a human egg cell contained a fully-formed miniature person, who would begin growing after fertilisation.

LINKED ACTIVITIES

1. **Dramatise refraction.** Three students should link arms in a straight line abreast, to represent a packet of light travelling through the air. A straight chalk line drawn on the floor represents the edge of the water. The 'packet of light' should approach the 'water' at an acute angle, taking big steps to represent high speed. As each person crosses the chalk line they should take smaller steps, to represent slower speed. *(Their direction will change as they enter the 'water' because one person will slow down before the others.)*



2. **Make a projector.** Hold a photographic slide very close to a bright light bulb. Hold a convex lens in front of the slide and move it forward and back until the image of the slide appears on a screen located 2–3 m away. *(The image appears because the light is bent and focussed as it moves through the lens.)*



CREATIVE LITERACY Before the development of microscopes, most people believed that living things could grow out of inorganic matter. For example, fleas could be produced from dust. Write how you would convince these people that such spontaneous generation does not happen.

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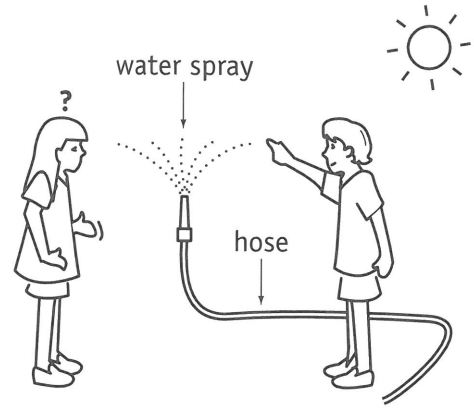
62 RAINBOW

You can make a rainbow with a garden hose.

HOW TO DO IT

This activity works best in direct sunlight, and in an outside area with a plain, dark surface.

1. Adjust the nozzle on a garden hose so it is producing a fine spray of water.
2. Direct the spray between two students — one who is directly facing the sun and another who is facing away from the sun. *(The student who is facing away from the sun will see a rainbow in the spray, but the other student will not.)*



EXPLANATION

Sunlight enters a water droplet and is refracted, causing it to be broken up into the colours of the spectrum. The light is also internally reflected from the back of the droplet, so the student who is facing the sun is standing 'behind the mirror' and doesn't see this reflected light.

Dogs are colour-blind, so their world looks like a black and white movie.

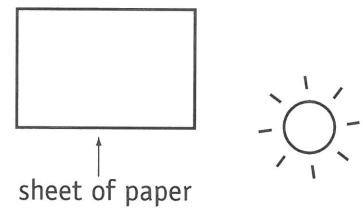
Bulls are colour-blind. A bullfighter has a red cape, but the bull responds to the movement of the cape rather than its colour.

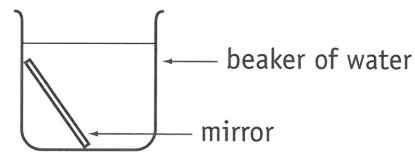
A laser is a beam of light of one pure colour, yet lasers can be made powerful enough to vaporise any substance, including diamond and steel.

CREATIVE LITERACY Imagine that you and your friends are a wave of light approaching a sun-shower. Accurately describe what happens to you as you enter the rain.

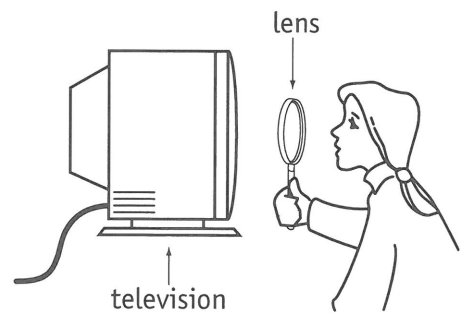
LINKED ACTIVITIES

1. **Model a raindrop.** Stand a pocket mirror in a beaker of water in direct sunlight. Hold a sheet of paper at least 1 m from the beaker in a position to catch the sunlight that has been reflected from the mirror. *(The colours of the spectrum will appear on the paper.)*





2. **Watch TV.** Put a colour video picture on 'pause' and look at the screen with a magnifying glass. How are black and white produced in a colour TV? *(The colours on the screen are made up from tiny red, green and blue blocks. All the colours will be glowing on a white part of the picture, but none of the colours will be glowing on a black part.)*



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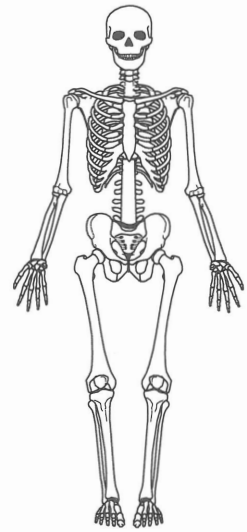
63 TAIL BONES

If you look at a human skeleton you will find it has a small tail.



HOW TO DO IT

Examine a full sized model of a human skeleton, and propose the function of each of the parts. *(There will be one small group of bones that does not have a function — the tail bones, located at the base of the spine.)*

EXPLANATION

The human tailbone (coccyx) is a vestigial structure (i.e., an organ that has degenerated because it is no longer necessary for survival). Humans have more than 100 other vestigial structures, including the appendix, wisdom teeth, body hair, and ear muscles.

A six-week-old human embryo has a full tail, but it later degenerates.

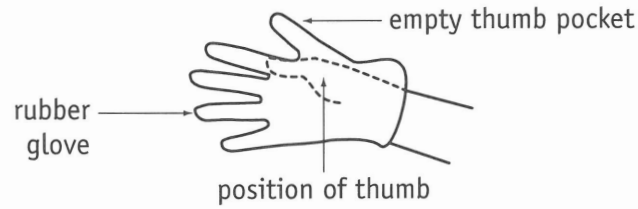
A giraffe has the same number of neck bones as a human.

The calcium in your bones was made in outer space. Like other heavier elements on earth, it was formed by fusion reactions inside stars and then released into space when the star exploded as a supernova, long before the earth was formed.

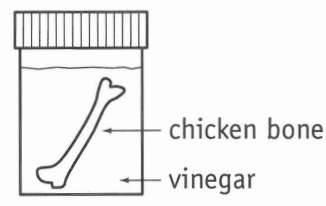
CREATIVE LITERACY Play skeletal celebrity heads. One student stands at the front of the class. The teacher writes the name of a bone on the blackboard behind him/her. The student has to ask yes/no questions of the class until he/she has enough clues to say the name of the bone.

LINKED ACTIVITIES

1 **Put your thumb out of action.** Insert your hand into a rubber kitchen glove but don't put your thumb in its hole. Instead, jam it under the rubber so it is immobilised, and you only have the use of your four fingers. Try to unwrap your lunch and eat a sandwich. *(Humans are the only animals with opposable thumbs. Without it, our manipulative ability is greatly reduced.)*



2 **Make a bendy bone.** Put a clean chicken bone in a jar of vinegar and let it soak for 1–2 weeks. *(Even though the bone still looks normal, it will be soft and pliable because the acid has broken down the calcium carbonate.)*



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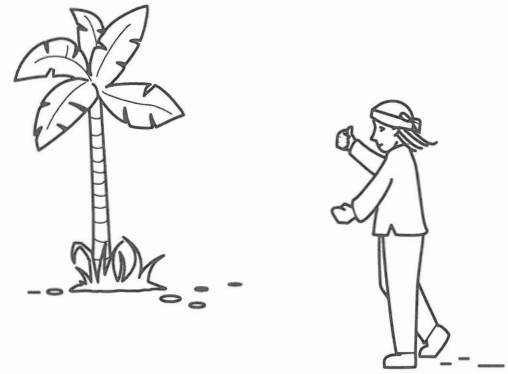
64 LEFT AND RIGHT

You may be right-handed, but you can still be left-eyed.

HOW TO DO IT



1 Note whether you are left-handed or right-handed as indicated by the hand you write with.

2 Close your left eye, hold your thumb out and align it with another object.

3 Open your left eye. (*If your thumb suddenly appears to be further to the right then your left eye is dominant. If it stays where it is then your right eye is dominant.*)

EXPLANATION

The left side of the brain controls the right side of the body, and vice versa. Also, different parts of the brain control different parts of the body, and some of our actions are dominated by one side or the other. About 90% of humans are right handed but only 70% are right eyed.

HISTORICAL NOTE

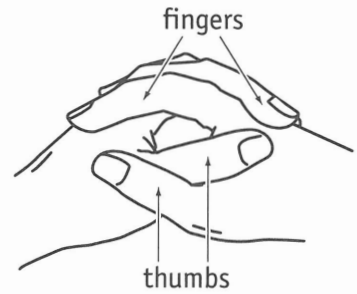
During the 1960s, people who exhibited aggressive or antisocial behaviour were sometimes given lobotomies — the top front of their brain was removed. This did not kill them but made them very 'passive'.

TO INVESTIGATE

Do all right-handed people fold their arms with their right arm on top?

Other animals can have dominant sides. Elephants are either left-tusk dominant or right-tusk dominant. You can tell by looking at the size of the tusk — the dominant tusk is the one that is shorter because it has worn away from constant use.

LINKED ACTIVITIES



1 **Are you right-thumbed or left-thumbed?** Put your hands together 'in prayer' with your fingers interlocked. (*If you are left-thumbed then your left thumb will be on top.*)

2 **Try to fool your brain.** Turn your wrists so your hands are back to back, then move them across each other and clasp your fingers together. Turn your hands in towards your body so your fingers are facing you. Ask your partner to point to one of your fingers, without touching it. Now try to move that finger. (*The fingers are back to front, so it will be difficult for your brain to sort out what the eyes are seeing and to select the correct finger.*)

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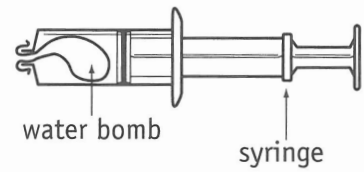
65 LUNG MODEL

You can inflate a balloon by remote control.

HOW TO DO IT

1 Cut the nozzle off a large plastic syringe, so that only a stump is left.
2 Push the plunger most of the way into the syringe.
3 Push the closed end of a water bomb through the nozzle hole and into the syringe.
4 Fold the neck of the water bomb over the stump of the nozzle.
5 Pull the plunger quickly and firmly. *(The water bomb will inflate inside the syringe.)*



EXPLANATION

When you breath in, your chest cavity becomes enlarged because the diaphragm contracts downwards and the ribs move outwards. This reduces the air pressure in the chest cavity, so the higher, ambient air pressure pushes air into the lungs. In the syringe model, the plunger represents the diaphragm.

When the human embryo is about 5 mm long it has gill slits instead of lungs. They are never used for breathing, but they are evidence that we have evolved from fishlike ancestors.

The inside surface of your lungs is larger than the floor of your classroom.

In a crowded cinema you could normally expect to be inhaling 60,000 microbes (bacteria and spores) with each breath you take.

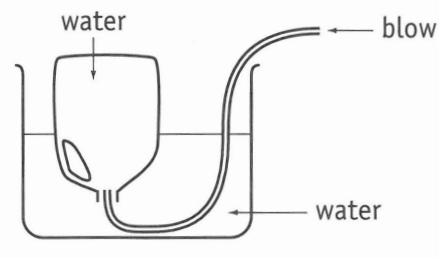
Frogs can breathe through their skin.

We should be glad that insects don't have lungs. All insects are small because they have a relatively inefficient way of bringing oxygen to their inner tissues. If they had lungs, we could have praying mantises the size of giraffes.

In China alone, smoking causes 2,000 deaths every day.

LINKED ACTIVITY

Measure your vital capacity. Take a 4 L plastic orange juice bottle, fill it with water and invert it in a large tub that also contains some water. Insert a rubber tube into the neck of the bottle. Take a very deep breath and try to empty your lungs into the bottle. Mark the new water level on the bottle. *(The vital capacity is the maximum amount of air you can expel from your lungs. It is not your lung capacity because you cannot completely empty your lungs.)*



CREATIVE LITERACY Imagine you are a particle of oxygen. Describe your experiences as you enter someone's lungs, and afterwards.

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66 HANDS UP

If you put your hand up it will become paler.

HOW TO DO IT

1 Face away from your partner. Tell your partner to put one hand up, as if answering a question in class, but leave the other hand on the desk.

2 After 10 seconds, quickly examine both of your partner's hands together. (*You will be able to identify the high hand because it will be noticeably paler.*)

EXPLANATION

Blood that flows upwards from the heart must work against the force of gravity. The result is a reduction in blood flow to the high hand, so it appears paler.

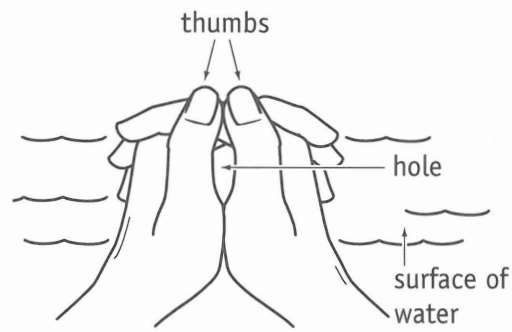
Soldiers on parade are told to wiggle their toes to prevent fainting. This muscular movement helps to push the blood up from the feet to the heart and brain.

Have you ever wondered why your faeces are always brown no matter what you have eaten? About 100,000 of your blood cells die each minute. Their haemoglobin is broken down to a yellow/brown substance, which is released into your intestine and voided with your waste food.

CREATIVE LITERACY On average, a blood cell returns to the heart about 45 seconds after leaving it. Imagine you are a blood cell; write a story entitled '45 seconds in the life of a blood cell'.

LINKED ACTIVITIES

1 **Model your heart.** Cup your hands together to make a chamber between them, with a hole at the top between your two thumbs. Immerse your hands in a tub of water, let some water enter the chamber through the bottom, then suddenly squeeze your hands together so a jet of water squirts through the hole between your thumbs. For how long can you maintain a consistent, accurate jet of water to a height of at least 20 cm at a rate of 70 per minute? *(Your hands will become fatigued and lose some rhythm after two or three minutes, but your heart muscle must do this for your whole life.)*



2 **Compare your fitness.** Find the pulse in your wrist and measure your resting pulse rate (number of beats per minute). After some mild exercise, measure your pulse rate again every 2 minutes until it returns to normal. Present this information on a graph. *(When you exercise, your muscles need more oxygen so your pulse rate speeds up. The fitter you are the lower your resting pulse rate, the lower your pulse rate during exercise, and the faster your pulse rate returns to normal after exercise.)*

HISTORICAL NOTE

Bloodsucking leeches were used to treat headaches during the 19th century. A standard treatment was to place several specimens of *Hirudo medicinalis*, which can grow up to 10 cm long, on the patient's temples and let them suck blood.

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67 EATING BISCUITS

How many biscuits can you eat in one minute?

HOW TO DO IT

1. Predict how many cracker biscuits you can eat in one minute, then remove that number of biscuits from the box and stack them on the table in front of you.
2. Note the time, then start eating them one at a time. You must swallow the previous one before you start a new one. *(Most people will only finish three in one minute.)*

EXPLANATION

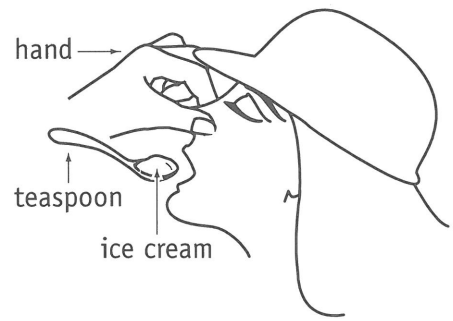
One of the functions of saliva is to moisten food so that it can be easily swallowed. Dry cracker biscuits absorb a lot of saliva, so as your mouth dries out you will find it more difficult to swallow.

Some animals swallow their food twice. When food is scarce, rabbits eat droppings straight from their anus. This allows them to get maximum use out of their food.

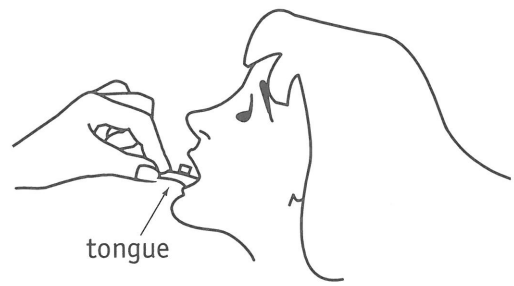
Vampire bats drink blood. They are active at night and their saliva contains an anaesthetic so their sleeping prey (usually farm animals) do not feel the bats' teeth entering their skin. These bats, which live in Central America, need to drink about half their weight in blood each night just to survive.

LINKED ACTIVITIES

1. **Taste some ice cream.** Ask your partner to close his/her eyes. Now bring a tub of Neapolitan ice cream into the room. Pinch your partner's nose tightly closed, then use a teaspoon to put a small amount of one flavour into his/her mouth. Your partner should try to name the flavour. Present the flavours in random order. *(Your partner will probably be able to identify the flavours, but with difficulty. The tongue tastes only sweet, sour, salty and bitter. More subtle flavours are picked up by our sense of smell. When your nose is blocked, it is difficult to discriminate between foods that have similar textures.)*



2. **Hold your tongue.** Put a small piece of bread on your tongue, then hold the tip of your tongue with your fingers. Try to swallow the bread. *(You will be unable to swallow until the tongue can push the food towards the back of the mouth.)*



HISTORICAL NOTE

Archaeologists can get a lot of information by studying teeth. In 1927, Davidson Black accurately described Peking man after examining one single tooth that had been found.

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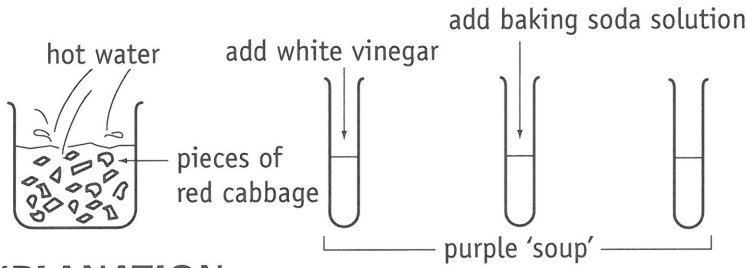
68 RED CABBAGE INDICATOR

A purple 'soup' will change to red (if one clear liquid is added) or to green (if another clear liquid is added).

HOW TO DO IT

Safety glasses and laboratory coats should be worn.

1. Fill a beaker with small pieces of red cabbage, then pour hot tap water into the beaker and mix it until the water turns purple.
2. Pour a small amount of the purple liquid into each of three test tubes.
3. Add white vinegar (a weak acid) to the first test tube. (*The solution will turn red.*)
4. Add baking soda solution (a weak base) to the second. (*The solution will turn green.*)
5. Leave the third test tube as a control.



HISTORICAL NOTE

Scurvy is a debilitating disease caused by lack of vitamin C. Up until about 1750, British sailors were more likely to die from scurvy than from enemy attack. On long voyages, a captain could confidently predict that half his crew would become bedridden by scurvy.

EXPLANATION

The pigment in red cabbage can be used as an acid/base indicator. However, the physiological reason for this is not known.

TO INVESTIGATE

What would happen if the red and green solutions above were mixed? (*The solution will turn purple as the acid and base neutralise each other.*)

Raw cabbage is therapeutic. Nursing mothers who suffer from mastitis (breast infections) can be cured by placing cabbage leaves in their bras.

Some acid rain has been measured at pH 1.5, which is more acidic than lemon juice.

Our stomach contains hydrochloric acid, with a pH of about 1. It is no wonder your throat seems to burn when you vomit!

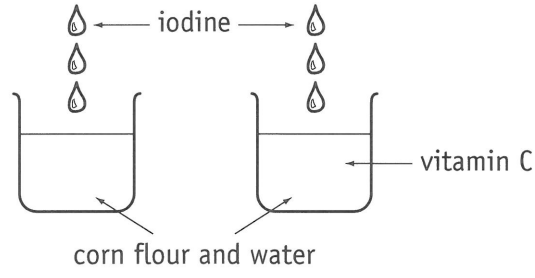
Vitamin C is an acid (ascorbic acid). Weight for weight there is more vitamin C in a green pepper than in an orange.

There is a microscopic alga, called *Cyanidium*, which can grow in concentrated sulfuric acid.

The planet Venus has an atmosphere with clouds that look similar to those on earth. However, its clouds consist of a mixture of carbon dioxide and sulfuric acid vapour.

LINKED ACTIVITY

Test for ascorbic acid. Mix 1/4 teaspoon of cornflour with 100 mL of water, then split it into two small beakers. Add a crushed vitamin C tablet to the first beaker and stir it well. Put 3 drops of iodine solution into each beaker. *(The iodine will turn black in the second beaker but not the first.)*



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69 PHENOLPHTHALEIN

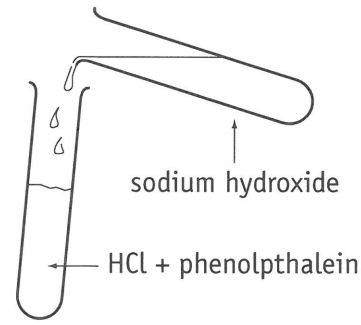
When one clear liquid is poured into another clear liquid a red colour will appear and then disappear.

HOW TO DO IT

Safety glasses and laboratory coats should be worn.

1 Put 1–2 cm of dilute hydrochloric acid in a test tube and add a squirt of phenolphthalein.

2 Add a few drops of dilute sodium hydroxide. (*A red colour will appear and then disappear.*)



EXPLANATION

The red colour appears where the phenolphthalein is in contact with the sodium hydroxide (phenolphthalein is colourless in acid or neutral conditions, but turns red when it is in the presence of a strong base). The red colour disappears again as the base mixes with the acid and becomes neutralised.

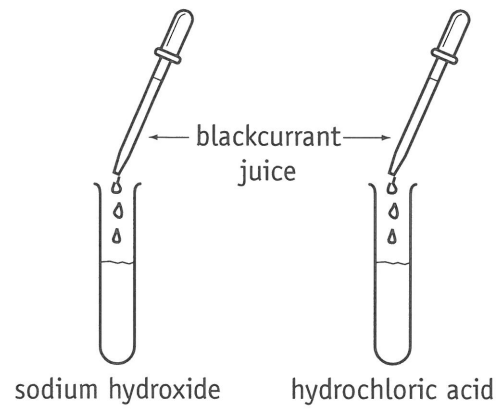
Phenolphthalein is also a laxative. It irritates the intestine and stimulates peristalsis.

Ant and bee stings contain acid, so they can be neutralised by the application of a weak base, such as baking soda.

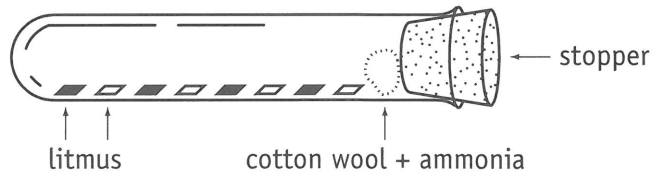
CREATIVE LITERACY Write the word phenolphthalein down the page, one letter on each line. On each line, write a fact about acids and bases starting with that letter.

LINKED ACTIVITIES

1 **Which brand of blackcurrant juice is a good indicator for bases?** *Safety glasses and laboratory coats should be worn.* Put a small amount of dilute sodium hydroxide in one test tube and a small amount of dilute hydrochloric acid in another. Add a few drops of your chosen blackcurrant juice to each. *(A good indicator for bases will turn green in the first test tube, but will remain red in the second.)*



2 **Use litmus paper to detect vapours.** *TEACHER DEMONSTRATION ONLY — This activity involves ammonia and should be carried out in a fume cupboard.* Tear off four pieces of red litmus and four pieces of blue litmus, each about 5 mm wide. Lay a large test tube on its side and use a glass rod to push the litmus pieces into the test tube. Arrange them so they are in a line, evenly spaced, and with the colours alternating. Put a few drops of ammonia on a small piece of cotton wool and insert it into the opening. Seal the test tube with a cork. *(The ammonia fumes will diffuse down the tube, turning the red litmus blue, but not changing the blue litmus.)*



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70 BURN MAGNESIUM

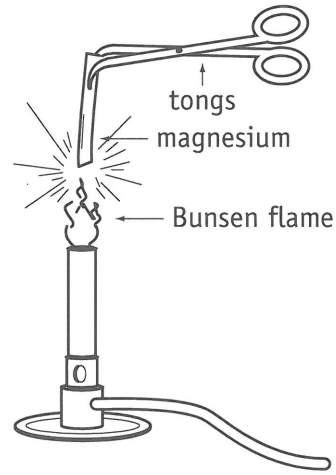
You can set fire to metal.

HOW TO DO IT

Safety warning: don't stare directly at the burning magnesium. Safety glasses and laboratory coats should be worn.

1 Break off a ribbon of magnesium about 3 cm long.

2 Hold one end with a pair of tongs and insert the other end into a Bunsen flame until it ignites. *(The magnesium will burn with a dazzling light.)* Hold the burning magnesium above an evaporating basin to collect the ash.



EXPLANATION

Magnesium burns with a very bright light as it combines with oxygen to form grey powdery magnesium oxide.

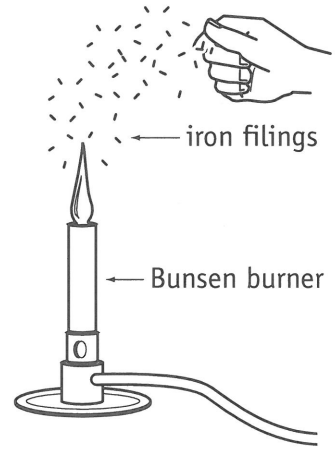
TO INVESTIGATE

Does magnesium weigh more or less after you burn it?

Fireworks use magnesium. The colours in fireworks come from other substances that are added. Strontium and calcium salts give a red colour, barium compounds produce green, and copper gives blue and green.

Rust is heavier than iron. This is because rust is iron oxide: it contains the same amount of iron, as well as some oxygen.

CREATIVE LITERACY Debate the statement 'Combustion is the most important chemical reaction'.

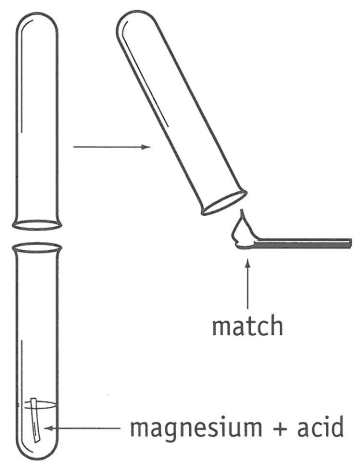


LINKED ACTIVITIES

1 **Burn iron.** *Safety glasses and laboratory coats should be worn.* Sprinkle some iron filings into a blue Bunsen flame. *(The metal will burn in a fountain of sparks. Small particles burn more readily because the surface area of the material is increased.)*

2 **Burn hydrogen.** *TEACHER DEMONSTRATION ONLY Safety glasses and laboratory coat should be worn.* Put about 2 cm of dilute hydrochloric acid in a test tube and add a 2 cm length of magnesium. Position a second test tube to collect the gas, then hold a lighted match to its opening. *(Hydrogen is lighter than air, so it collects in the top test tube. The gas will explode with a 'pop' as it reacts with oxygen to form water.)*

About 75% of all the matter in the universe is hydrogen. It is one of the main components of stars, including our sun.



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71 BURNING KEROSENE

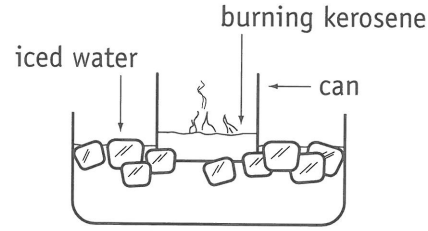
You can stop kerosene burning by making it colder.

HOW TO DO IT

TEACHER DEMONSTRATION ONLY
Safety glasses and laboratory coat should be worn. Students should stand well clear of the burning kerosene.

1 Put some kerosene in a small can and use a Bunsen flame to light it.

2 Use a pair of tongs to hold the base of the can in icewater. (*The flame will go out.*)



EXPLANATION

When the can is placed in the icewater the kerosene begins to lose heat. When it drops below its ignition temperature combustion ceases.

Some materials explode when they reach ignition temperature. Interestingly, the ingredients of gunpowder are found in hot springs (sulfur), wood fires (charcoal) and compost heaps (saltpetre).

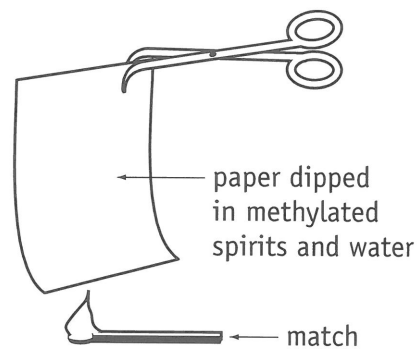
CREATIVE LITERACY Scientists believe that the burning of fuels contributes to global warming. However, unlike other western countries, Australia has not agreed to reduce its production of greenhouse gases. Write a letter to the editor expressing your opinion about this.

HISTORICAL NOTE

Alfred Nobel, after whom the peace prize is named, actually devoted much of his research to the development of explosives. One accidental explosion killed his brother.

LINKED ACTIVITIES

1 **Make flames that don't burn paper.**
TEACHER DEMONSTRATION ONLY
Safety glasses and laboratory coat should be worn. Make a 1:1 mixture of methylated spirits and water. Immerse a sheet of paper in the liquid then pull it out and hold it at the corner with a pair of tongs. Light the paper at the bottom. *(The sheet of paper will be covered in flames, which will eventually go out, leaving the paper intact. The water keeps the temperature of the paper low, so that it doesn't ignite.)*



2 **Solve the following problem.** Some brands of safety matches claim to be damp resistant. Design an experiment to test these claims, not forgetting to incorporate adequate safety measures into your method. Are they more water resistant than other brands? Can you design a simple, cheap way to waterproof a match? (*One way is to lightly coat it with candle wax.*)

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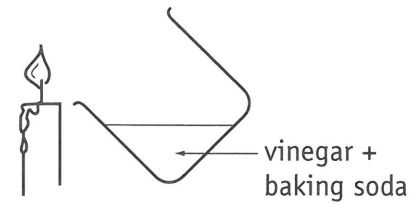
72 FIRE EXTINGUISHER

You can pour invisible gas over a candle flame, to put it out.

HOW TO DO IT

1 Put about 50 mL of vinegar in a 250 mL beaker and add a small, flat teaspoon of baking soda. Let the beaker stand as the reaction proceeds.

2 As the reaction dies down, hold the spout of the beaker next to the base of a candle flame (rather than above the flame) and tip the beaker slightly, being careful not to pour any of the liquid or bubbles out. (*The candle will go out.*)



EXPLANATION

Vinegar contains about 4% acetic acid. When this is mixed with baking soda (sodium bicarbonate) bubbles of carbon dioxide gas are produced. Carbon dioxide gas is heavier than air, so it sits in the beaker rather than floating away. When the wick becomes blanketed with carbon dioxide, oxygen cannot get to the flame and combustion ceases.

The air is only 0.03% carbon dioxide.

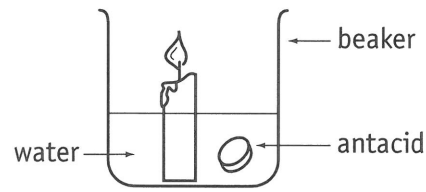
Some deaths that occur during bushfires are a result of asphyxiation rather than burning. This happens when the bushfire uses up so much oxygen that there is not enough for people to breathe.

HISTORICAL NOTE

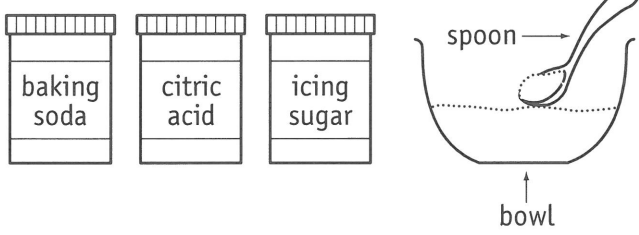
Joseph Priestly was a scientist who became famous for his work on carbon dioxide and its use in beverages. Interestingly, he lived next door to a brewery.

LINKED ACTIVITIES

1 **Fire extinguisher II.** Take a short piece of candle and attach its base to the bottom of a beaker. Pour some water into the beaker, then light the candle. Put an antacid tablet in the water. (*The candle will go out as it becomes blanketed with carbon dioxide.*)



2 **Make sherbet.** Mix 1 teaspoon of baking soda and 1 teaspoon of citric acid with several tablespoons of icing sugar. Taste your mixture. *(Saliva dissolves the acid and carbonate so they can react together to produce carbon dioxide, which tastes 'fizzy' on your tongue.)*



CREATIVE LITERACY Create your own recipe for 'designer sherbet' by using other ingredients such as baking powder (rather than baking soda), jelly crystals, antacid powder etc. Write an advertisement explaining why your sherbet is superior.

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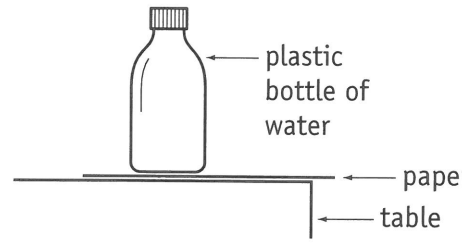
73 TABLECLOTH TRICK

You can pull a sheet of paper out from under a bottle of water.

HOW TO DO IT

1 Stand a plastic soft drink bottle, full of water, on top of a sheet of paper.

2 Quickly jerk the paper out from under the bottle. *(The bottle will remain where it was.)*



EXPLANATION

The inertia of the bottle resists any change. The more quickly you pull the paper, the greater the inertia. When you pull the paper very quickly then the inertia is sufficient to overcome the friction between the paper and the bottle.

TO INVESTIGATE

What would happen if you used an empty plastic bottle? What would happen if you used a tablecloth instead of paper?

CREATIVE LITERACY Write a patter for a magician performing the tablecloth trick.

Moving objects also have inertia. Icebergs move when the wind blows them. But when the wind drops, their inertia will keep them moving for hours.

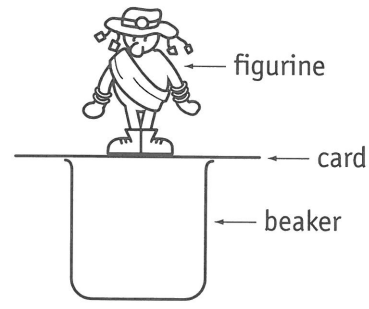
Inertia can kill you. The passengers in a moving car have inertia, so they will tend to keep moving forward even if the car suddenly stops. Cars have claimed 30 million lives worldwide since they were first invented.

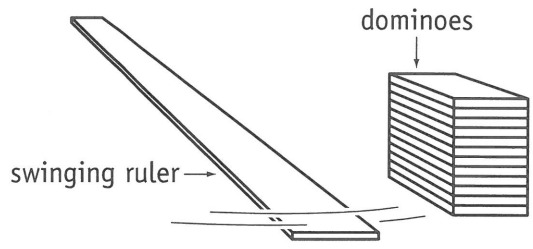
Spacecraft have inertia. The Voyager probes were launched in 1977 and are presently headed into deep space at about 15 km/s with no propulsion. They carry time capsules containing pictures, sounds, music and language from earth in case of an encounter with extraterrestrials.

LINKED ACTIVITIES

1 **Drop a model into a beaker.** Lay a card on top of a beaker, then stand a plastic figurine on top of the card. Flick the card quickly. *(The inertia of the figurine prevents it moving with the card, so it will drop into the beaker.)*

2 **Remove a domino.** Make a stack of about a dozen dominoes on a smooth table top. Firmly hit the bottom one with the edge of a ruler. *(Inertia keeps the rest of the stack intact, so only the bottom one will be ejected.)*





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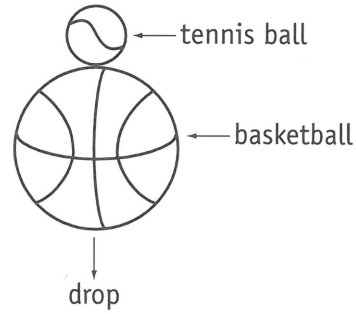
74 BASKETBALL AND TENNIS BALL

A basketball can be used to propel a tennis ball.



HOW TO DO IT

This activity should be done outside in a clear area with a hard, flat surface.

1. Hold a tennis ball on top of a basketball at about chest height.
2. Drop the two balls, still in contact with each other, onto a hard, flat surface. (*The basketball will not bounce much, but the tennis ball will shoot into the air.*)

EXPLANATION

The momentum from the large ball is transferred to the small ball. The small ball has less mass so, in order to conserve momentum, its velocity increases.

TO INVESTIGATE

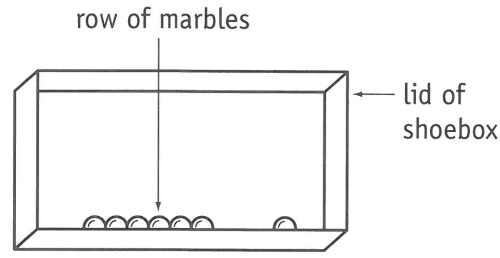
What would happen if the positions of the two balls were reversed? What would happen if you used two balls of roughly equal size but different weights? (Try a golf ball and a ping-pong ball.)

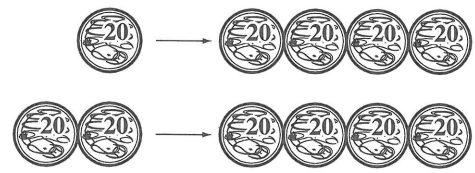
The earth regularly has collisions. About 500 meteorites hit the ground every year. The largest known meteorite weighs 61 tonnes.

The Himalayas were caused by a collision between India and Asia. This collision is still happening, and the Himalayas are getting higher every year.

LINKED ACTIVITIES

1. **Play with marbles.** Put six marbles in the lid of a shoebox and tilt it so the marbles are in a row along one edge. Make sure they are all touching. Roll another marble along the edge so it collides with the end of the row. *(The marble at the other end will roll away.)* Roll two marbles together to collide with the end of the row. *(Two marbles will move from the other end.)*
2. **Play with coins.** Repeat the activity above but use a straight row of 20c coins on a smooth tabletop.
3. **Solve the following problem.** Devise a way to drop an egg from a second storey window so that it doesn't break. *(This can be achieved by using a parachute to reduce the velocity of the egg and by using crumpled paper to reduce the rate of the collision.)*





CREATIVE LITERACY Many cartoon characters are involved in amazing collisions. Describe one example you believe to be scientifically inaccurate. Explain your reasons.

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75 BOTTLE ROCKET

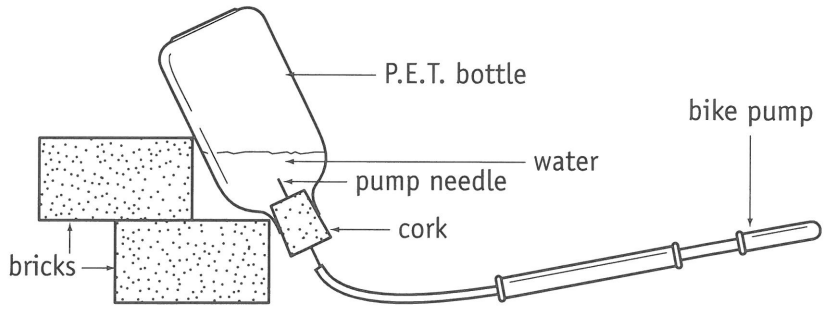
You can use a bicycle pump to launch a plastic bottle.

HOW TO DO IT

TEACHER DEMONSTRATION ONLY

This activity needs to be done outside, in an open area. Safety warning: stand clear of the front of the bottle.

1. Pour water into a plastic bottle until it is about 1/8 full.
2. Insert a pump needle through a cork and fit it into the neck of the bottle.
3. Invert the bottle and attach a bicycle pump to the needle.
4. Pump air into the bottle until it launches. *(It should travel at least 10 m.)*



EXPLANATION

The change in momentum of the water equals the change in momentum of the bottle. The bottle has less mass than the water, so it moves in the opposite direction at a relatively greater velocity.

CREATIVE LITERACY Many Hollywood movies have gunfights in which the person who gets shot is knocked backwards several feet, while the shooter hardly experiences any recoil. Find an example of this from television and explain why it is scientifically inaccurate.

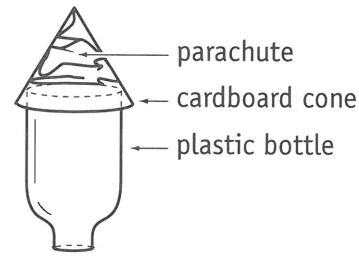
TO INVESTIGATE

What would happen if the bottle was 3/4 full of water? What would happen if it was empty of water?

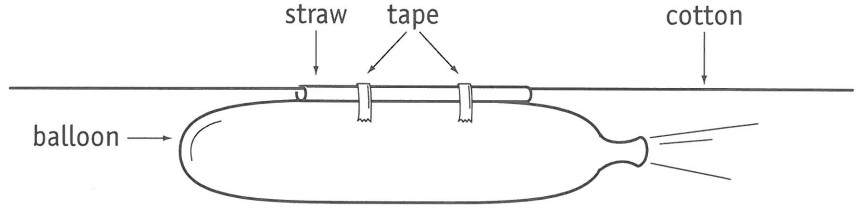
A garden hose has jet propulsion. If you suddenly turn the water on full force then the nozzle springs back and moves around on the ground. Jet boats are powered in a similar way.

LINKED ACTIVITIES

1. **Solve the following problem.** Develop a way to attach a plastic-bag parachute to the bottle rocket so that it will open as the bottle begins to descend (*One solution is to attach it under a loosely-fitted cardboard nose-cone*).



2. **Solve the following problem.** Design a 2-balloon rocket that can travel a maximum distance along a length of string (i.e., at least as long as a classroom). Use a packet of assorted shaped balloons. *(A drinking straw can be used to attach a balloon to the string. Sausage balloons work well.)*



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76 CANISTER ROCKET

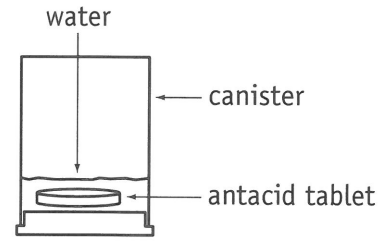
You can make a photo canister blast off.

HOW TO DO IT

TEACHER DEMONSTRATION ONLY

Safety warning: stand well clear of the canister.

1. Select a plastic photo canister in which the cap fits inside rather than over the outside. The cap should 'snap' on.
2. Add water until the canister is about ¼ full.
3. Add an antacid tablet then quickly put the lid on and place the canister upside down on the floor. Stand clear. *(After a few moments, the canister will shoot upwards with a 'pop'.)*



EXPLANATION

Carbon dioxide gas builds up until the gas pressure exceeds the frictional force between the canister and its lid. Momentum is conserved during the take-off — as the canister has much less mass than the lid/earth system, it moves into the air at a higher velocity.

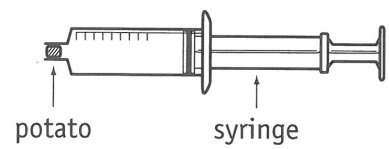
Animals developed jet propulsion long before humans. Squid and octopus are propelled by squirting water from a funnel near their tentacles. Their ancestors were doing this long before humans appeared on the planet.

Airports are dangerous places for birds because they can be sucked into jet engines. In fact, modern jet engines are constructed so that they can 'inhale' a seagull without losing any power.

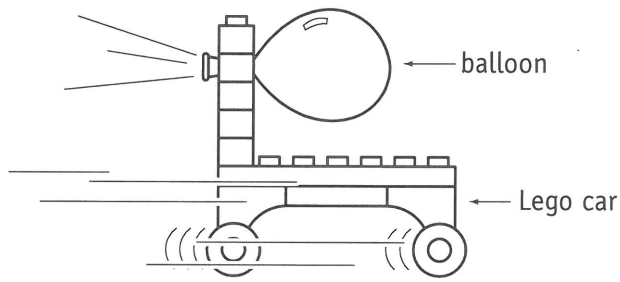
Blowpipes are traditional weapons in parts of Malaysia. Using only their lung power, hunters can blow a dart over 100 m.

LINKED ACTIVITIES

1. **Shoot a piece of potato.** *TEACHER DEMONSTRATION ONLY — Safety glasses should be worn.* Take a large, plastic syringe and pull back the plunger. Insert the nozzle into a disk of raw potato so a small piece is wedged inside the nozzle. Aim it away from other people and quickly depress the plunger.



2. **Solve the following problem.** Use Lego to make a balloon-powered car. What is the maximum distance you can get it to travel?



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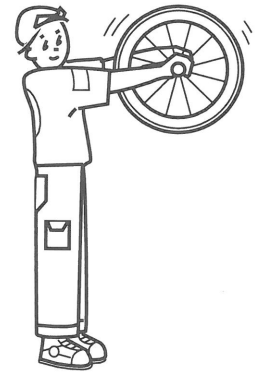
77 SPINNING WHEEL

A spinning bicycle wheel seems to have a mind of its own.



HOW TO DO IT

1. Remove the front wheel from an adult's bicycle.
2. Hold it in front of you and use pliers (if necessary) to grip the end of each axle.
3. Ask another person to spin the wheel vigorously.
4. While the wheel is spinning quickly, turn to face another direction. *(You will feel the wheel resisting the change in direction.)*

EXPLANATION

The spinning wheel has inertia that maintains the direction in which it is aligned. To change the direction of the spinning wheel you need to overcome this inertia.

TO INVESTIGATE

What happens if you don't spin the wheel as quickly? What happens if you use a smaller wheel?

CREATIVE LITERACY Imagine that you have to teach Einstein to ride a bicycle. Write a list of instructions using as much scientific jargon as possible.

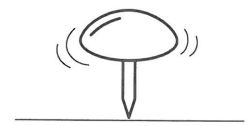
A bicycle will remain upright as long as its wheels are spinning quickly. When a bicycle is put on rollers and pedalled quickly it still remains upright even though it is not moving forward at all.

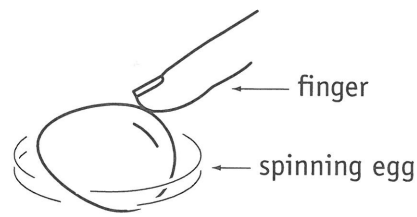
HISTORICAL NOTE

The earth's spin is slowing down. About 400 million years ago, when fish were beginning to populate the oceans, the earth was spinning faster, so there were about 400 days in a year. This has been calculated from historical records of the timing and visibility of eclipses.

LINKED ACTIVITIES

1. **Spin a drawing pin.** Hold the spike of a drawing pin. Give it a sharp twist and let it drop to the table top. *(Its inertia will make it remain upright as long as it is spinning strongly.)*
2. **Use inertia to tell if an egg is raw or hard boiled.** Place a hard boiled egg on a table and spin it. Firmly touch your finger to the top of the egg to stop it spinning, then quickly let it go again. *(The egg will remain still.)* Now repeat the process using a raw egg. *(The egg will resume spinning again when you let it go. The contents of a raw egg are semi-liquid, and their inertia makes them continue to spin slightly after the outer shell is stopped.)*





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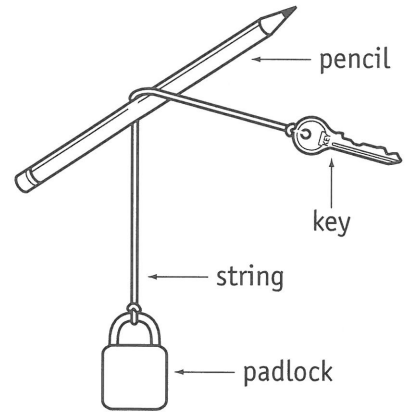
78 FALLING PADLOCK

When you let the key go, the padlock will fall towards the floor but will not hit it.



HOW TO DO IT

1. Obtain a length of string about 1 m long.
2. Tie one end to a heavy padlock, and the other to a small key.
3. Hold the key, and use a pencil to support the string about halfway down its length, as shown.
4. Let go of the key. *(The padlock will not hit the floor.)*

EXPLANATION

When you release the key, it begins to move in a circular path, like a pendulum. However, the length of its string is continually decreasing because the padlock is pulling it, so the key keeps speeding up. Its velocity increases so much that it wraps the string around the pencil.

TO INVESTIGATE

What would happen if the key was held closer to the pencil, or if the positions of the key and the padlock were reversed?

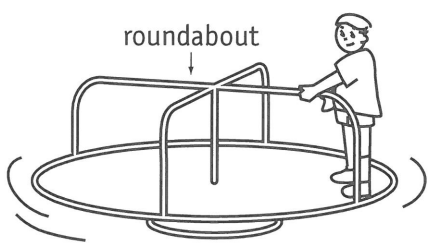
Creative Literacy Watch a video of Olympic figure skaters. Write a short report describing how they increase their speed when they go into a fast spin.

Pulsars are rapidly spinning stars. They are extremely small, having a diameter of only 20 km or so, and they spin at up to 642 times per second. It has been estimated that there are about 1 million pulsars in our galaxy, the Milky Way.

In the time it takes you to jump into the air and land again, the ground underneath you has moved over 200 m eastward, due to the earth's spin. You don't notice it because you have moved too.

LINKED ACTIVITIES

1. **Spin a chair.** Sit in an office chair that spins. Start the chair spinning with your legs stretched out in front of you, then draw them in close to your body. *(The speed will increase as you draw your legs in closer.)*
2. **Spin a roundabout.** Go to a playground. Stand near the edge of a roundabout and start it moving slowly. Now move to the centre of the roundabout and hold on tightly. *(The speed will increase when you move to the centre.)*



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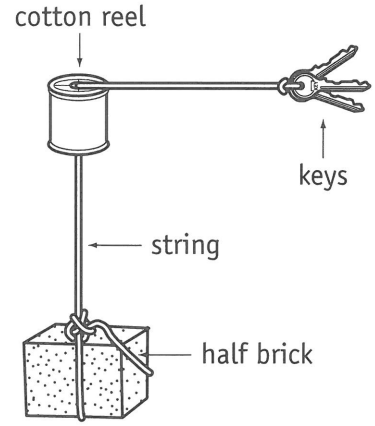
79 ROTATING KEYS

Three keys can lift a half brick.

HOW TO DO IT

TEACHER DEMONSTRATION ONLY — This activity should be done outside in a position where students can observe without risk of injury from the fast moving keys.

1. Cut a 1 m length of strong string and securely tie three big keys to one end.
2. Thread the rest of the string through the hole in a cotton reel and tie the other end to a small half brick.
3. Hold the cotton reel and use it to very vigorously spin the keys in a circle. *Safety warning: hold the cotton reel above your head so the keys won't hit you in the face. (The brick will be lifted off the ground.)*



EXPLANATION

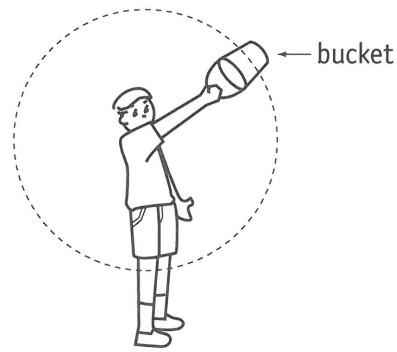
The force you have given to the keys would actually make them move in a straight line away from you if the string was cut. (Due to their inertia, they would resist a change in speed and direction and thus tend to move at a tangent to the spinning circle.) However, the inward-acting centripetal force exerted by the brick, via the string, prevents this from happening. When you spin the keys very quickly, the inertia overcomes the centripetal force, and the brick is lifted.

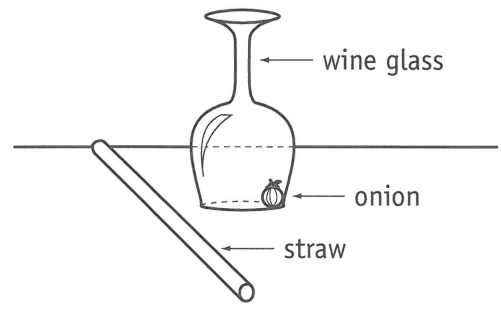
On the planet Venus, a day is longer than a year. Venus spins very slowly on its axis, so it can complete an orbit of the sun (one year) before it has completed a single rotation (one day).

There is a lot of disagreement about centrifugal force. Engineers use it in their calculations, but physicists say it doesn't even exist!

LINKED ACTIVITIES

1. **Swing a bucket of water.** This activity should be done outside. Get a small, plastic bucket and add water to it until it is about $^3/_4$ full. Begin swinging the bucket smoothly until you can swing it over your head in a big circle. *(If you swing it quickly the water will remain in the bucket.)*
2. **Try this bar trick.** Place a straw on the table and a small cocktail onion to one side of it. The problem is: how can you use an upside down wineglass to lift the onion over the straw to the other side? *(It can be done by placing the wineglass over the onion, then swirling the glass so the onion rolls around the inside surface. If you can get the onion moving quickly enough, it will rise up the inside of the wineglass until it is rolling around the widest point. The glass can then be lifted over the straw to deposit the onion on the other side.)*





CREATIVE LITERACY Describe a fairground ride involving centripetal force.

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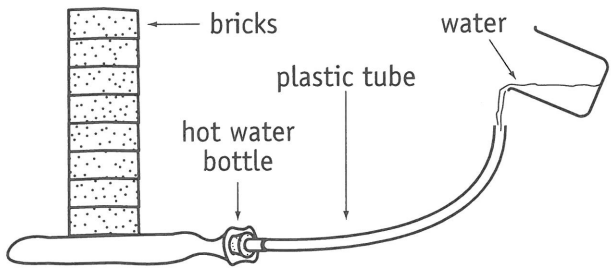
80 LIFTING BRICKS

A cup of water will lift a pile of bricks.

HOW TO DO IT

TEACHER DEMONSTRATION ONLY

1. Obtain a cork which has a short piece of glass tubing inserted through it and fit the cork into the opening of a hot water bottle.
2. Attach about 0.5 m of plastic tubing to the glass tube and make sure all the fittings are secure.
3. Stack a column of 8–10 bricks on top of the hot water bottle. *Safety warning: stay well clear of the bricks.*
4. Pour 200–300 mL of water into the tube. *(The bricks will fall over.)*



CREATIVE LITERACY Write a full description and explanation of the 'To Investigate' activity in narrative form and containing NO scientific terms.

EXPLANATION

The weight of the water in the plastic tube exerts pressure on the water in the hot water bottle. This pressure is multiplied over the whole inner surface of the hot water bottle and is enough to lift the bricks.

HISTORICAL NOTE

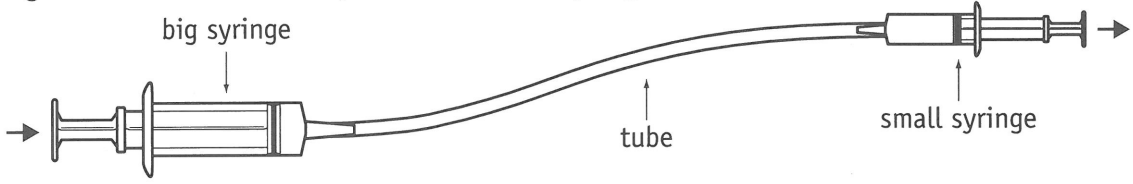
During World War II most submarines operated at very shallow depths — only 100 m or a bit more — but the average depth of the ocean is 3,700 m.

TO INVESTIGATE

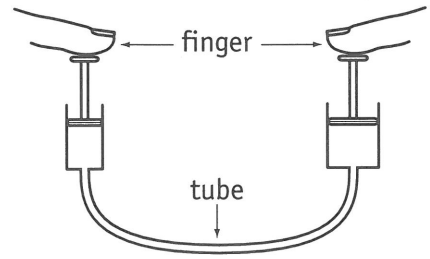
Does air act in the same way as water? Ask one of your friends to sit on the hot water bottle in place of the bricks.

LINKED ACTIVITIES

1. **Make a hidden syringe.** Cut a thin, flexible piece of plastic tubing at least 50 cm long. Attach a small syringe to one end and a large syringe to the other. Fill the syringes and tubing with water. Display the small syringe on the desk, but have the other syringe out of sight. Press the plunger on the hidden syringe. *(The plunger on the visible syringe will move outwards, seemingly of its own accord, as the pressure is transmitted through the fluid. The narrower piston will move a greater distance, but the change in volume will be exactly the same in both syringes.)*



2. **Feel the pressure exerted by the syringes above.** Place a finger on each plunger. Press downwards on the small plunger and try to resist the pressure on the big plunger. *(Your finger will be unable to resist the movement because the pressure is multiplied in the larger syringe.)*



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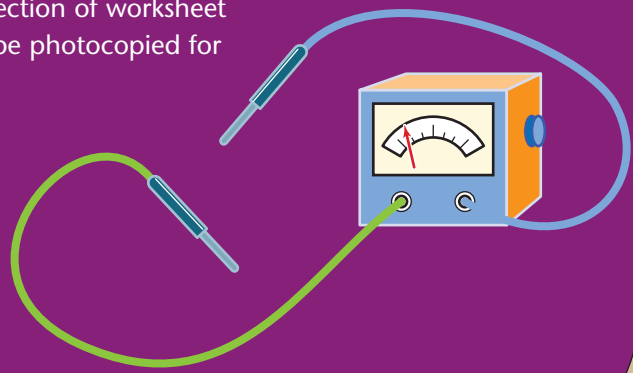

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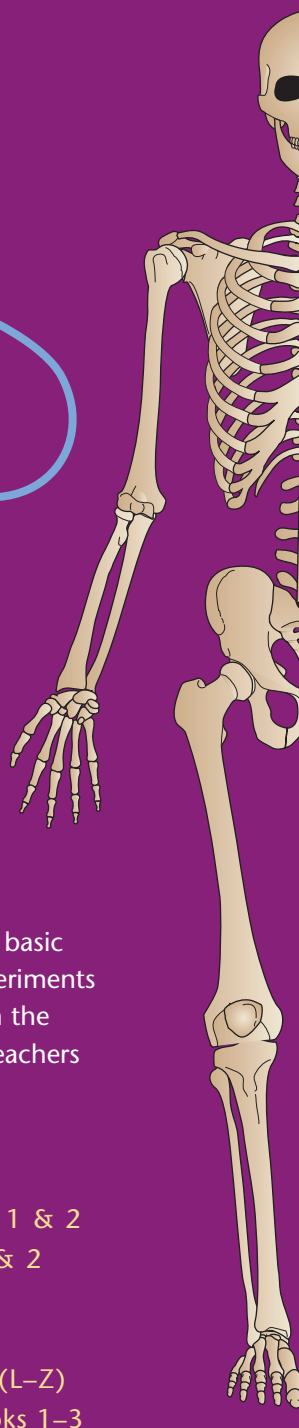

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